

ECE2150J Intro to Circuits

Chapter 7. First-Order Circuits

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7.1 Introduction

- Until now, we considered passive elements (R, L, and C), and an active element (op amp) individually.
- Now, we will see circuits with the combination of R, L, and C: Chapter 7: RC and RL; Chapter 8: RLC.
- **RC and RL circuits** are characterized by **first-order differential equations**. Hence, the circuits are collectively called **first-order circuits**.

Source of Energy

A circuit needs energy, either potential or current. In RL or RC circuits, there are two ways to provide energy to the circuits.

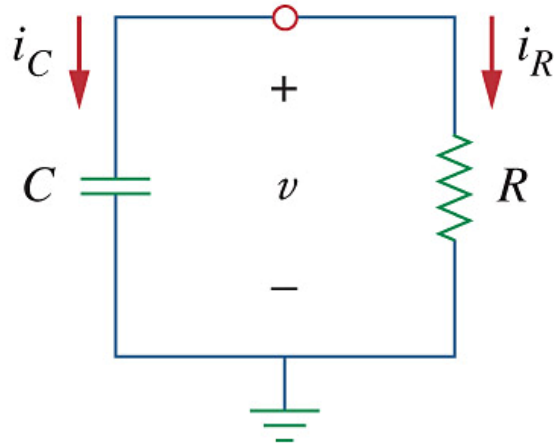
- (i) By energy **initially stored** in the capacitive or inductive element.
- (ii) By **independent sources**.

Source of Energy	Case (i)	Case (ii)
RC	7.2 The Source-Free <i>RC</i> Circuit: $v_c(t=0) = V_0$	7.5 An <i>RC</i> Circuit with Step Input
RL	7.3 The Source-Free <i>RL</i> Circuit: $i_L(t=0) = I_0$	7.6 An <i>RL</i> Circuit with Step Input

7.2 The Source-Free *RC* Circuit

- A source-free *RC* circuit occurs when its dc source is suddenly disconnected. The **energy already stored in the capacitor** is released to the resistor.

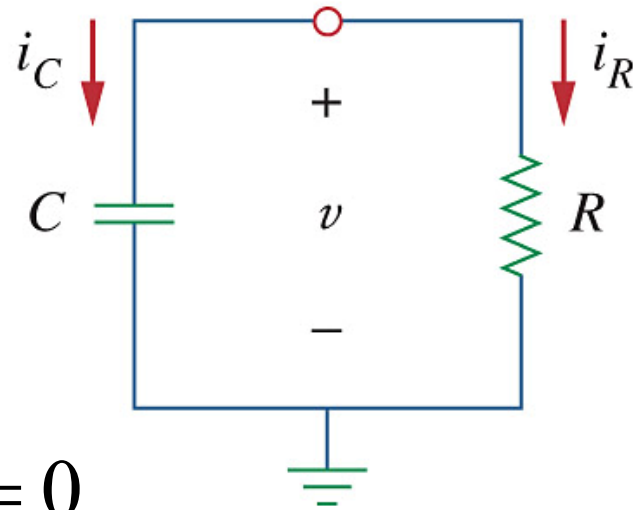
Source-Free RC Circuit Response



The capacitor is initially charged. We assume that at time $t=0$, the initial voltage is $v(0) = V_0$ with the corresponding value of energy stored:

$$w_C(0) = \frac{1}{2} C V_0^2$$

The circuit response, the capacitor voltage v in this case is



$$i_C + i_R = 0$$

$$i_C = C \frac{dv}{dt}, i_R = \frac{v}{R}$$

$$C \frac{dv}{dt} + \frac{v}{R} = 0 \quad \Rightarrow \quad \frac{dv}{dt} + \frac{v}{RC} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = 0$$

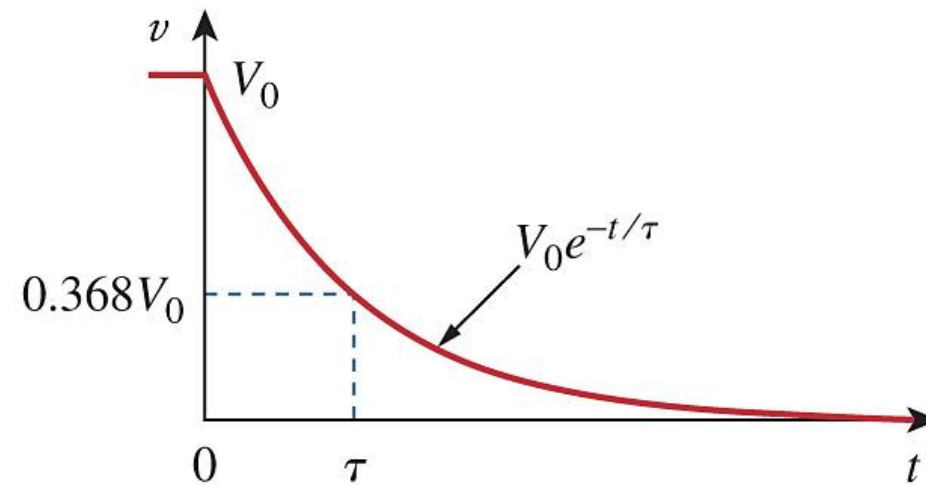
$$\frac{dv}{v} = -\frac{1}{RC} dt \text{ integrate both sides}$$

$$\ln v = -\frac{t}{RC} + C \rightarrow v = Ae^{-\frac{t}{RC}}$$

Using the initial condition $v(0) = V_0$, we get $v(t) = V_0 e^{-\frac{t}{RC}}$

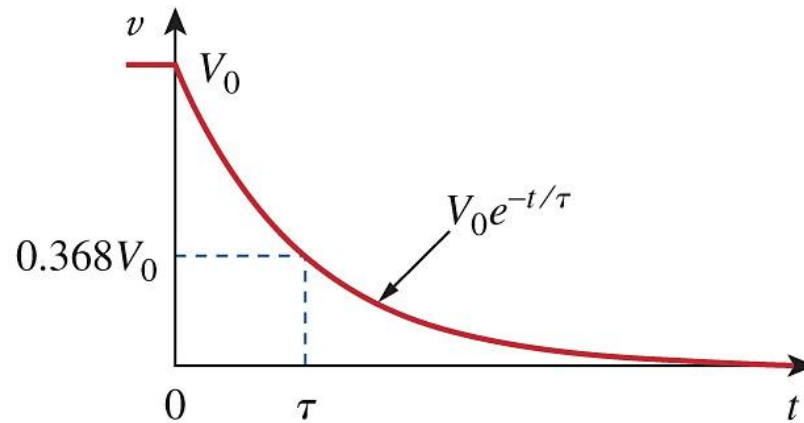
The response is due to the initial energy stored and the physical characteristics of the circuit. We call it **Natural Response, or zero-input**, of the circuit.

$$v(t) = V_0 e^{-\frac{t}{RC}}$$



The response of the source-free RC circuit: at $t=0$, $v=V_0$, and as t increases, v decreases toward zero. The rapidity with which v decreases is expressed in terms of **the time constant τ** .

Time constant τ



The time constant of a circuit is the time required for the response to decay to a factor of $1/e$ or 36.8% of its initial value. This implies that at $t = \tau$,

$$v = V_0 e^{-\frac{\tau}{RC}} = V_0 e^{-1}$$

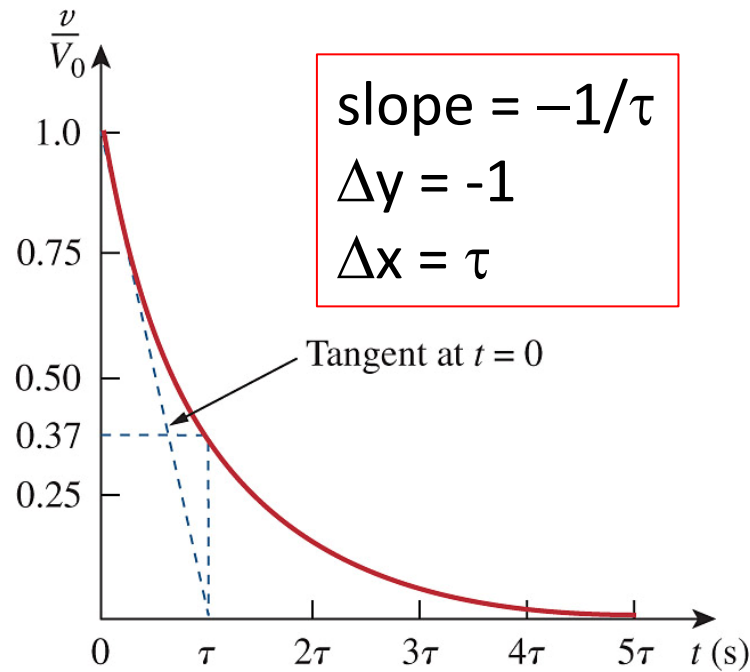
or

$$\tau = RC$$

The time constant may be viewed from another perspective. Evaluating the derivative of v/V_0 at $t = 0$, we obtain

$$v(t) = V_0 e^{-\frac{t}{\tau}}$$

$$\left. \frac{d}{dt} \left(\frac{v}{V_0} \right) \right|_{t=0} = -\frac{1}{\tau} e^{-t/\tau} \bigg|_{t=0} = -\frac{1}{\tau} \quad \text{slope at } t=0$$

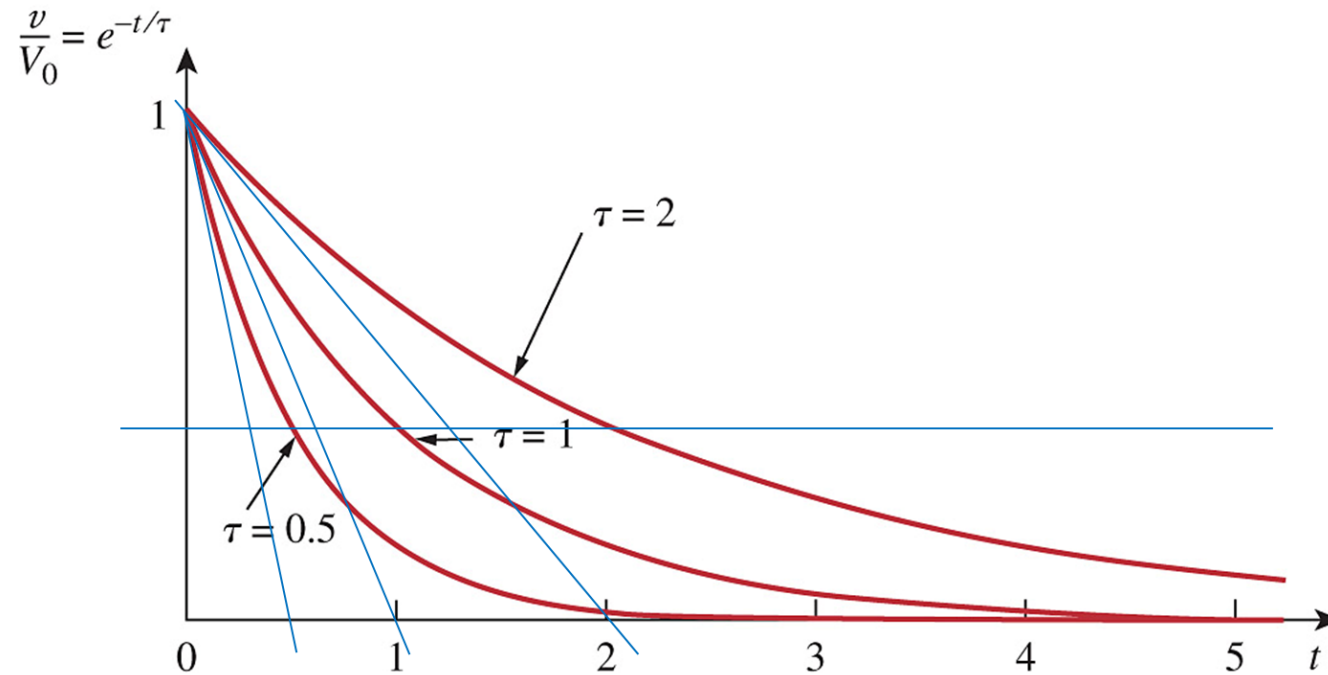


Thus, the time constant is the **initial rate of decay**. To find τ from the response curve, draw the tangent to the curve at $t = 0$. The tangent intercepts with the time axis at $t = \tau$.

TABLE 7.1 Values of $v / V_0 = e^{-t/\tau}$

t	τ	2τ	3τ	4τ	5τ
v / V_0	0.36788	0.13534	0.04979	0.01832	0.00674

After 5τ , v is less than 1% of the initial value V_0 . It is customary to assume that the capacitor is fully discharged after five time constants, i.e. the circuit reaches its **final state or steady state** when no changes take place with time.



The smaller the time constant, the more rapidly the voltage decreases, i.e. the faster response → quick dissipation of energy stored.

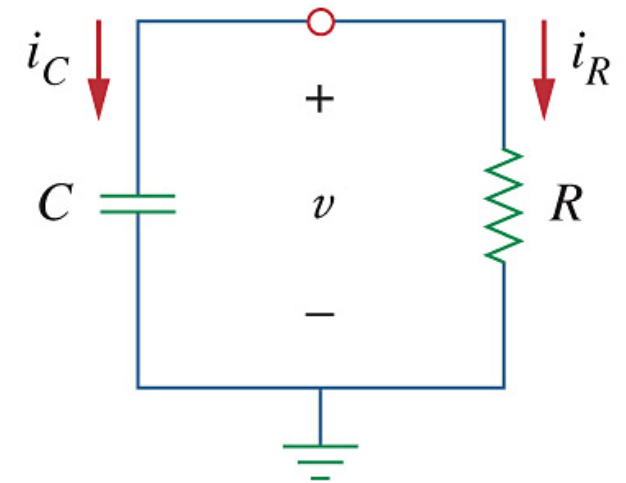
Circuit response at R

Go back to **the circuit response**, from the voltage in capacitor $v(t) = V_0 e^{-\frac{t}{RC}}$ we can find the resistor current

$$i_R = \frac{v}{R} = \frac{V_0}{R} e^{-t/\tau}$$

The power dissipated in the resistor is

$$p = vi_R = \frac{V_0^2}{R} e^{-2t/\tau}$$



The energy absorbed by the resistor up to time t is

$$\begin{aligned} w_R &= \int_0^t \left(\frac{V_0^2}{R} e^{-2t/\tau} \right) dt = \frac{V_0^2}{R} \frac{e^{-2t/\tau}}{-2/\tau} \Big|_0^t \\ &= \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau}) \end{aligned}$$

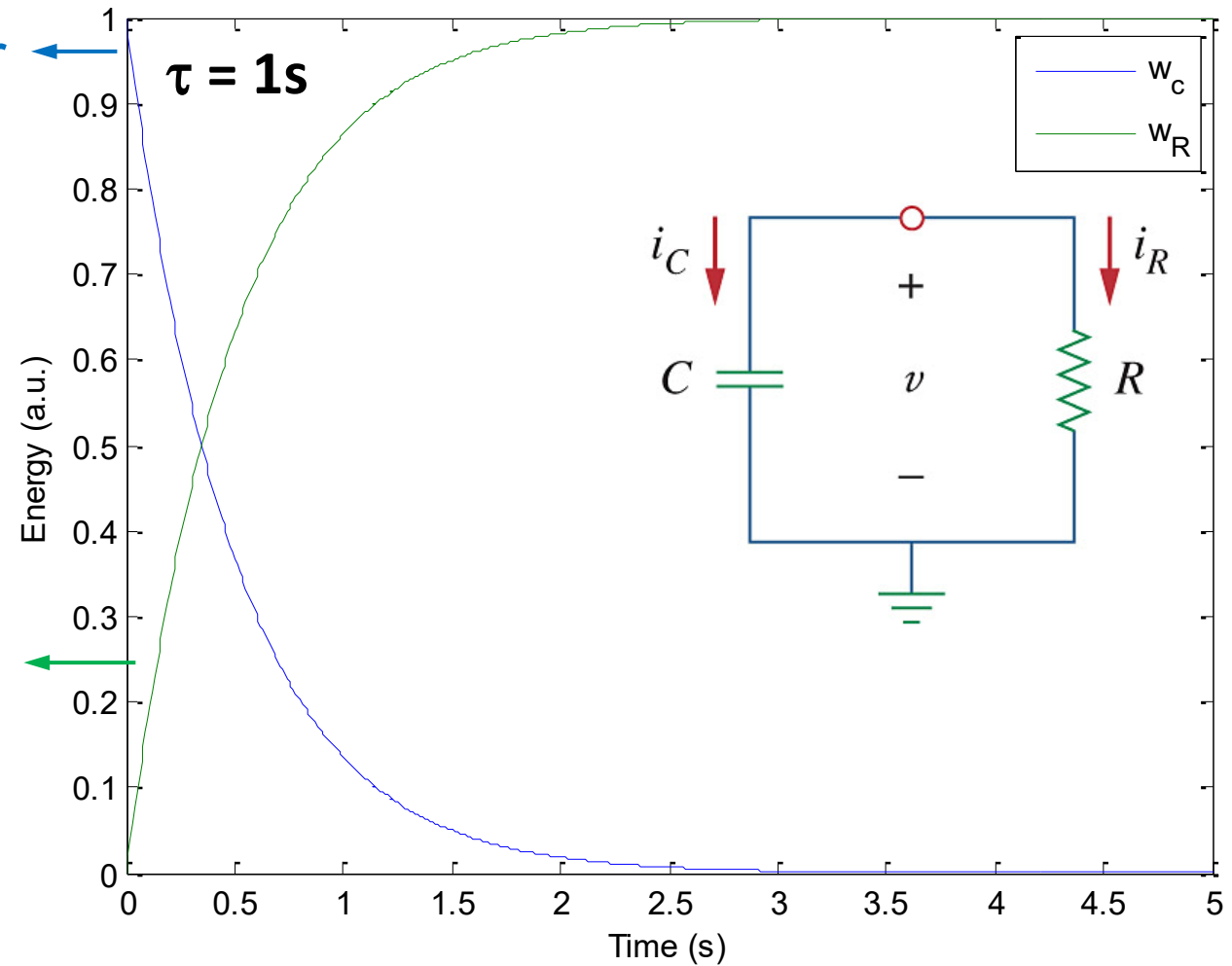
Notice that as $t \rightarrow \infty$, $w_R \rightarrow \frac{1}{2} C V_0^2 = w_C(0)$,

the energy initially stored in the capacitor

Energy in Circuit

Energy in Capacitor

Energy in Resistor



Summary: RC source free response

To find the RC source free response

1. The initial voltage $v(0) = V_0$ across the capacitor.
2. The time constant $\tau = RC$ where R is often the equivalent resistance at the terminals of C .

The response, the capacitor voltage V_C , is

$$V_C = V = V(0)e^{-t/\tau}$$

and then, we can find **other circuit responses**, V_R , i_R , I_C etc.

Practice Problem 7.1 Refer to the circuit in Fig. 7.7. Let $v_C(0) = 45$ V. Determine v_C , v_x , and i_o for $t \geq 0$.

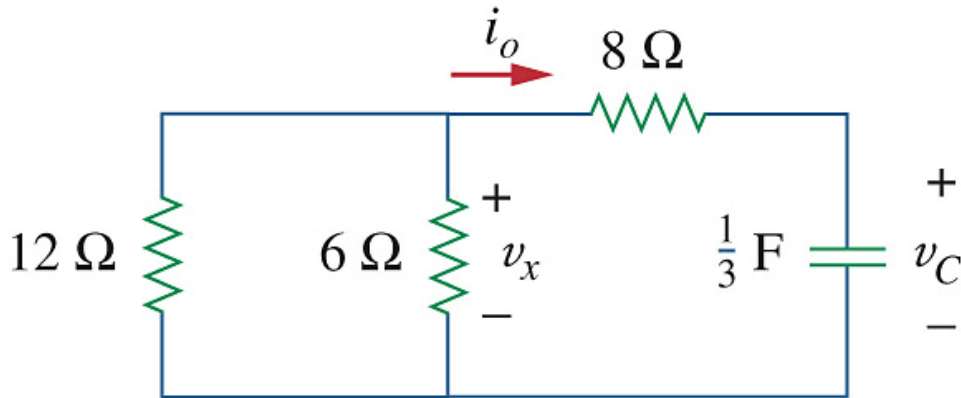


Figure 7.7 An RC circuit.

Solution :

The equivalent resistance seen at the terminals of the capacitor is

$$R_{eq} = 8 + 12 \parallel 6 = 12 \text{ } (\Omega)$$

The time constant is

$$\tau = R_{eq} C = 12 \times \frac{1}{3} = 4 \text{ (s)}$$

The capacitor voltage is

$$v_C = v_C(0)e^{-t/\tau} = 45e^{-t/4} = 45e^{-0.25t} \text{ (V)}$$

$$i_o = C \frac{dv_C}{dt} = \frac{1}{3} \times \frac{d}{dt} (45e^{-0.25t})$$

$$= 15 \times (-0.25e^{-t/4}) = -3.75e^{-t/4} \text{ (A)}$$

$$\begin{aligned} v_x &= 8i_o + v_C = 8 \times (-3.75e^{-0.25t}) + 45e^{-0.25t} \\ &= 15e^{-0.25t} \text{ (V)} \end{aligned}$$

Practice Problem 7.12 If the switch in Fig. 7.10 opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_C(0)$.

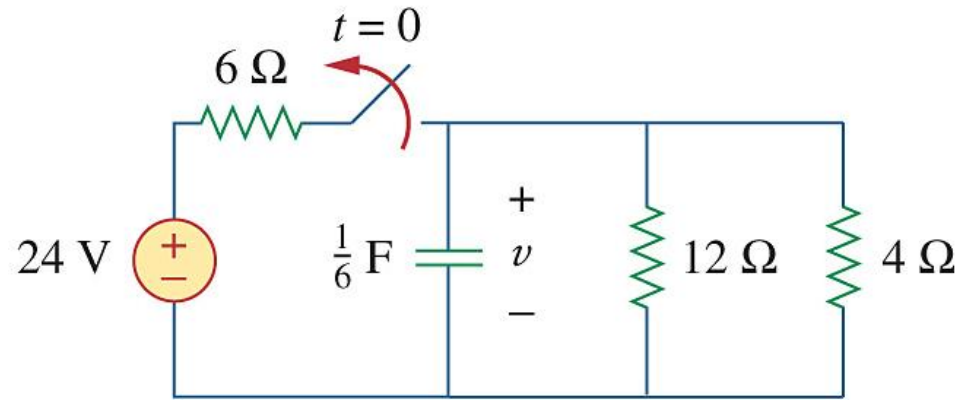
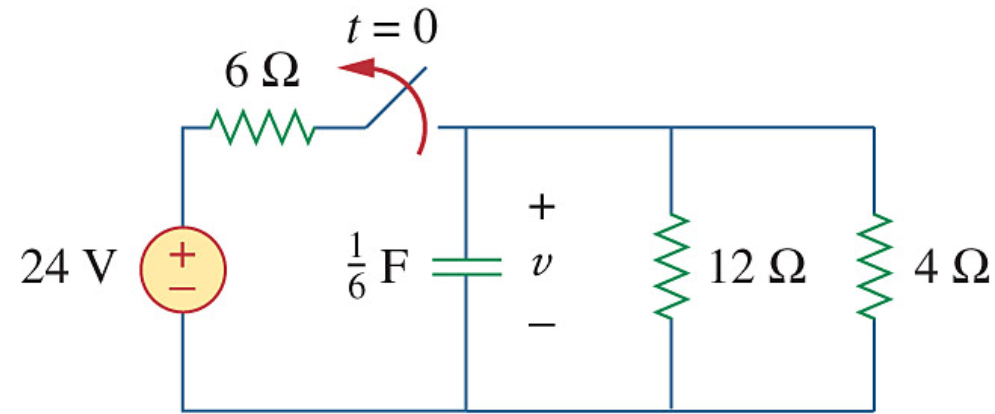


Figure 7.10



Solution :

When $t \leq 0$, the capacitor voltage

$$v(t) = 24 \times \frac{12 \parallel 4}{6 + 12 \parallel 4} = 8 \text{ (V)}$$

Hence, $v(0) = 8 \text{ V}$.

When $t \geq 0$, the circuit becomes a source-free RC circuit with

$$\tau = R_{eq} C = (12 \parallel 4) \times \frac{1}{6} = \frac{1}{2} \text{ (s)}$$

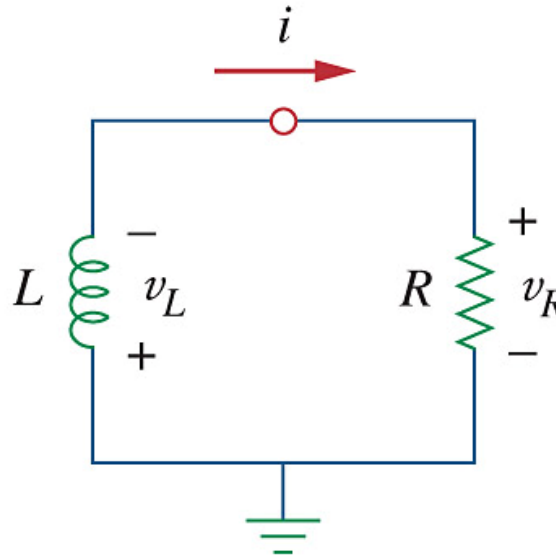
Therefore,

$$v(t) = v(0)e^{-t/\tau} = 8e^{-t/(1/2)} = 8e^{-2t} \text{ (V)}$$

$$w_C(0) = \frac{1}{2} C (v(0))^2 = \frac{1}{2} \times \frac{1}{6} \times 8^2$$

$$= \frac{16}{3} \approx 5.33 \text{ (J)}$$

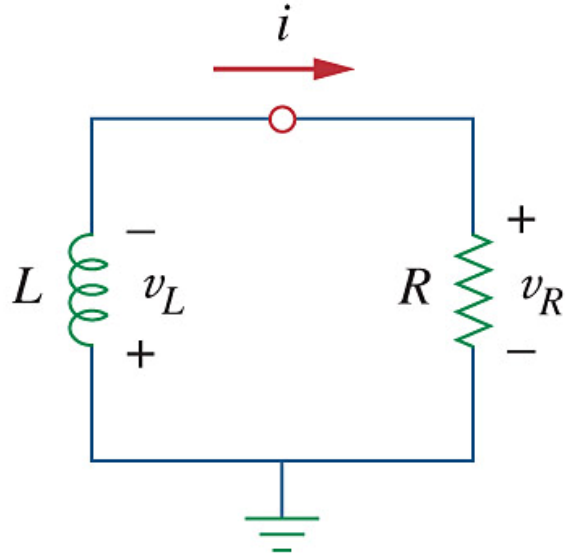
7.3 The Source-Free *RL* Circuit



Now, we think about the RL circuit with the initial current $i(0) = I_0$ through the inductor, and corresponding energy

$$w_L(0) = \frac{1}{2} L I_0^2$$

Source-Free RC Circuit Response



$$v_L + v_R = 0$$

$$v_L = L \frac{di}{dt}, v_R = iR$$

$$L \frac{di}{dt} + iR = 0 \quad \frac{di}{dt} + \frac{R}{L}i = 0$$

Solve the 1st order differential equation

$$i = Be^{rt} = Be^{-\frac{R}{L}t} = Be^{-\frac{t}{(L/R)}}$$

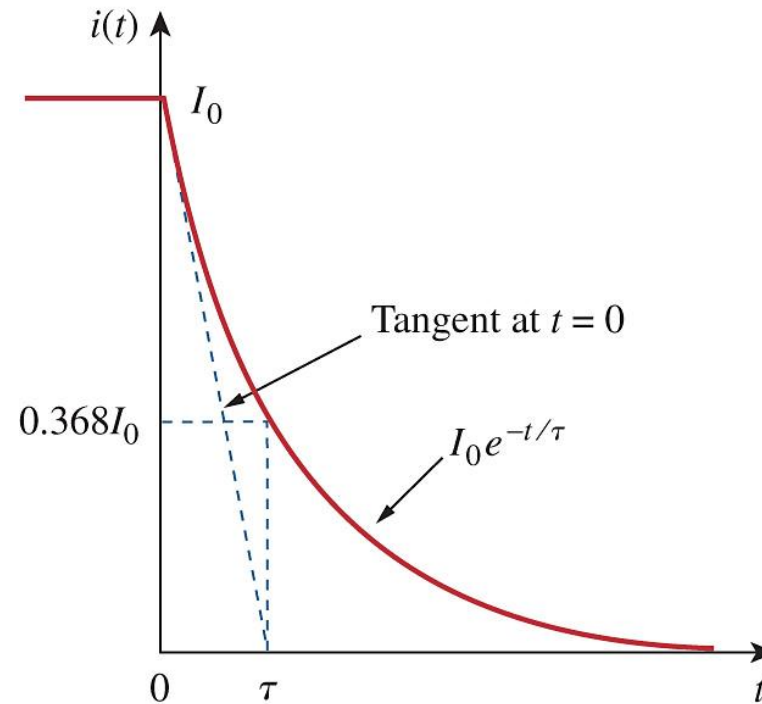
$$i(0) = B = I_0$$

$$i = I_0 e^{-\frac{t}{(L/R)}}$$

$$i = I_0 e^{-\frac{t}{L/R}}$$

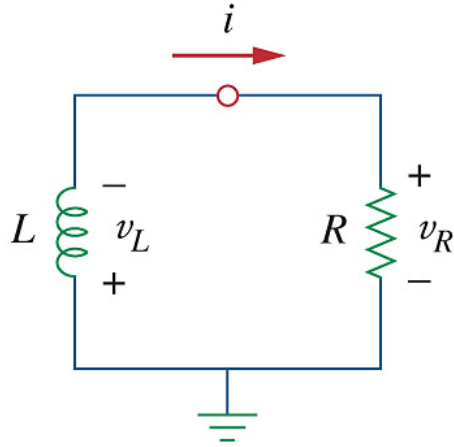
Let $\tau = L / R$, we have

$$i = I_0 e^{-t/\tau}$$



Because the response is due only to I_0 , it is called **natural response, or zero-input**, of the circuit.

Circuit response and Energy at R



Voltage at the resistor

$$v_R = iR = I_0 R e^{-t/\tau}$$

Power at the resistor

$$p = v_R i = I_0^2 R e^{-2t/\tau}$$

Energy at the resistor up to time t

$$w_R = \int_0^t \left(I_0^2 R e^{-2t/\tau} \right) dt = I_0^2 R \left. \frac{e^{-2t/\tau}}{-2/\tau} \right|_0^t$$

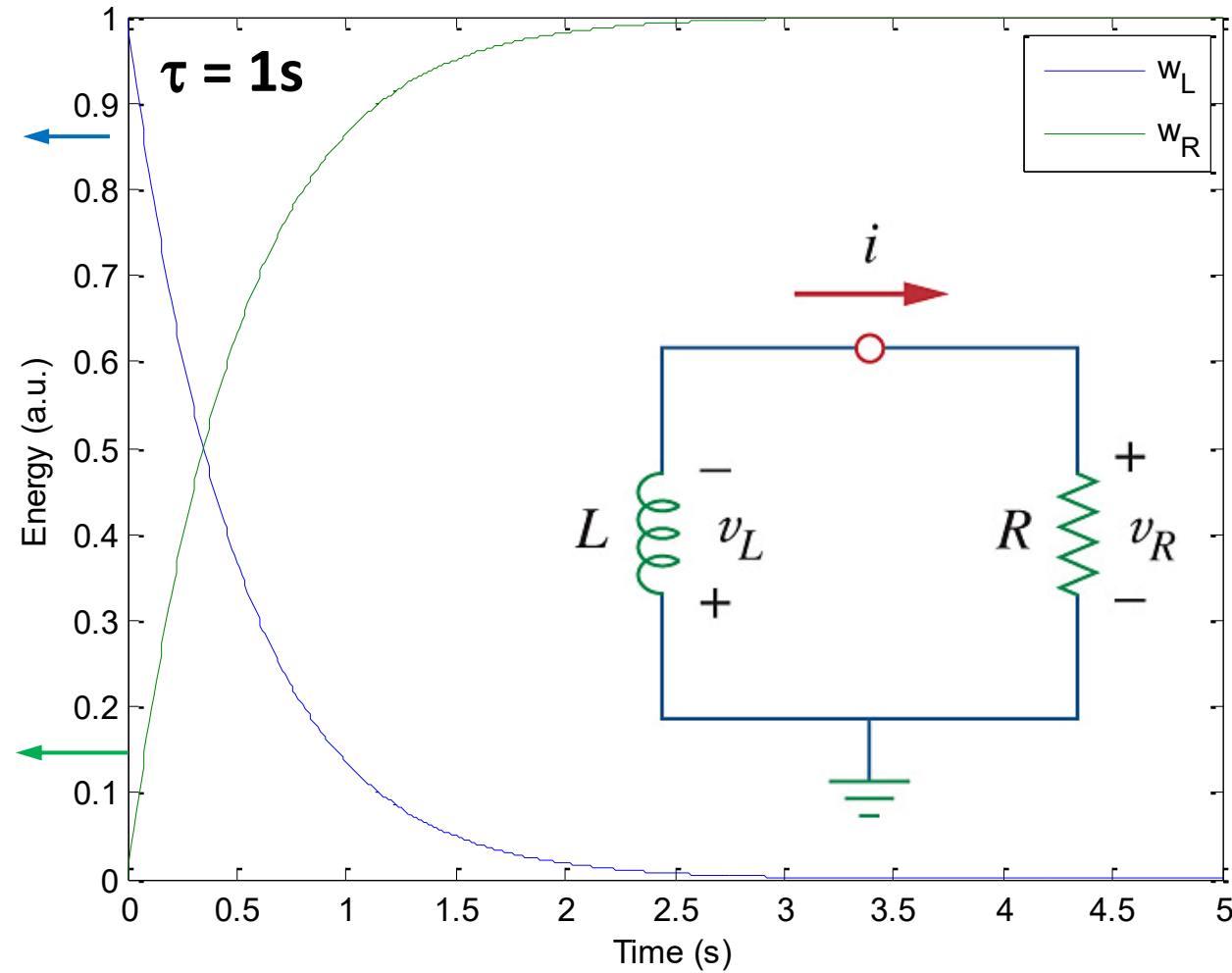
$$= \frac{1}{2} L I_0^2 \left(1 - e^{-2t/\tau} \right)$$

Notice that as $t \rightarrow \infty$, $w_R \rightarrow \frac{1}{2} L I_0^2 = w_L(0)$,
the energy initially stored in the inductor.

Energy in Circuit

Energy in Inductor

Energy in Resistor



Summary: RL source free response

To find the RL source free response

1. The initial current $i(0) = I_0$ through the inductor.
2. The time constant $\tau = L/R$ where R is often the equivalent resistance at the terminals of L .

The response, the inductor current i , is

$$i_L = i = i(0)e^{-t/\tau}$$

and then, we can find **other circuit responses**, V_L , V_R , i_R etc.

Practice Problem 7.4 For the circuit in Fig. 7.18, find $i(t)$ for $t > 0$.

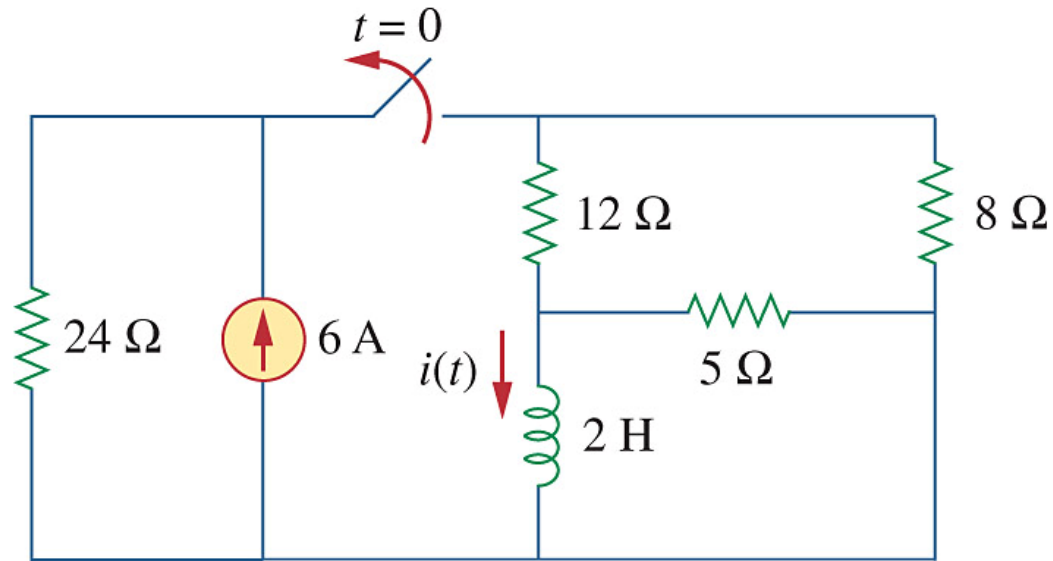
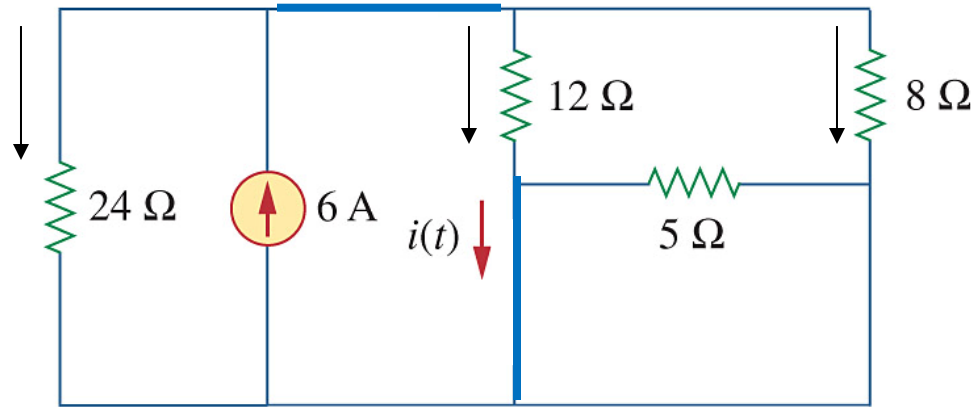


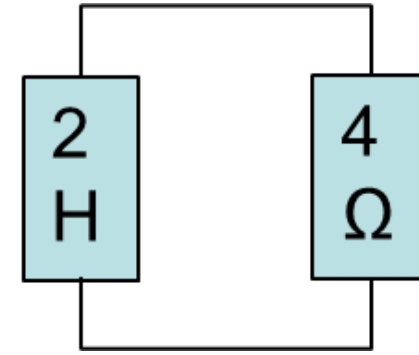
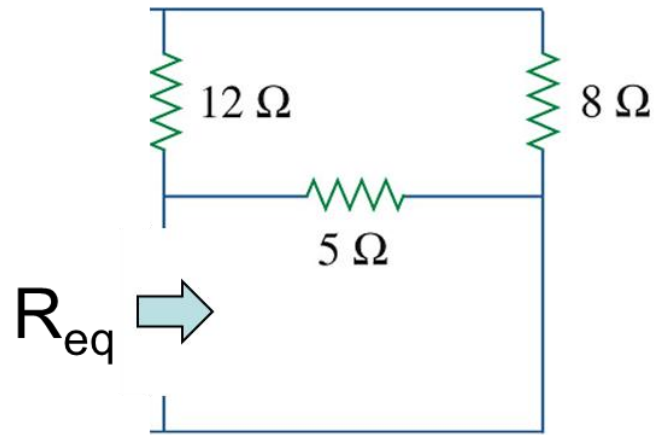
Figure 7.18



Solution :

When $t < 0$, the current through the 5-Ω resistor is zero.

$$i(t) = 6 \times \frac{24 \parallel 8}{24 \parallel 8 + 12} = 2 \text{ (A)}$$



When $t > 0$,

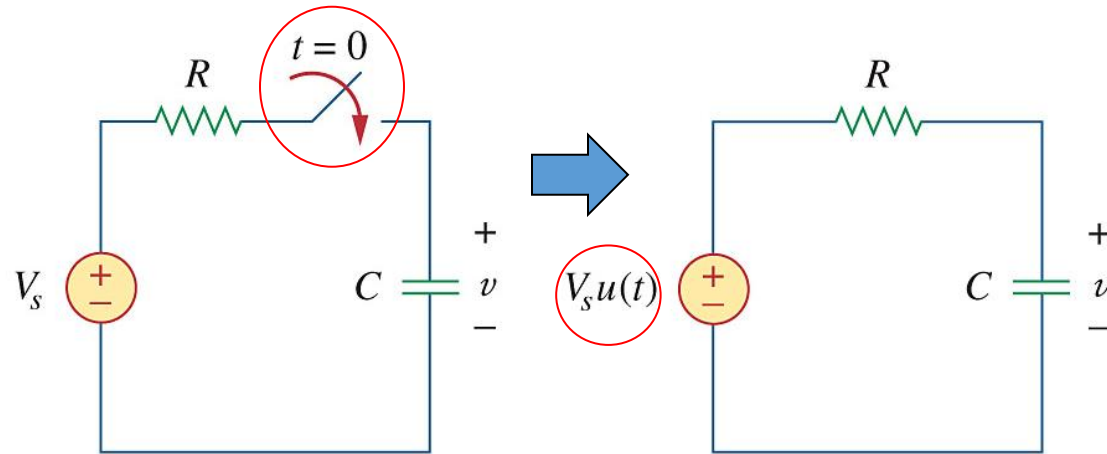
$$R_{eq} = 5 \parallel (12 + 8) = 4 \text{ } (\Omega)$$

$$\tau = L / R_{eq} = 2 / 4 = 0.5 \text{ (s)}$$

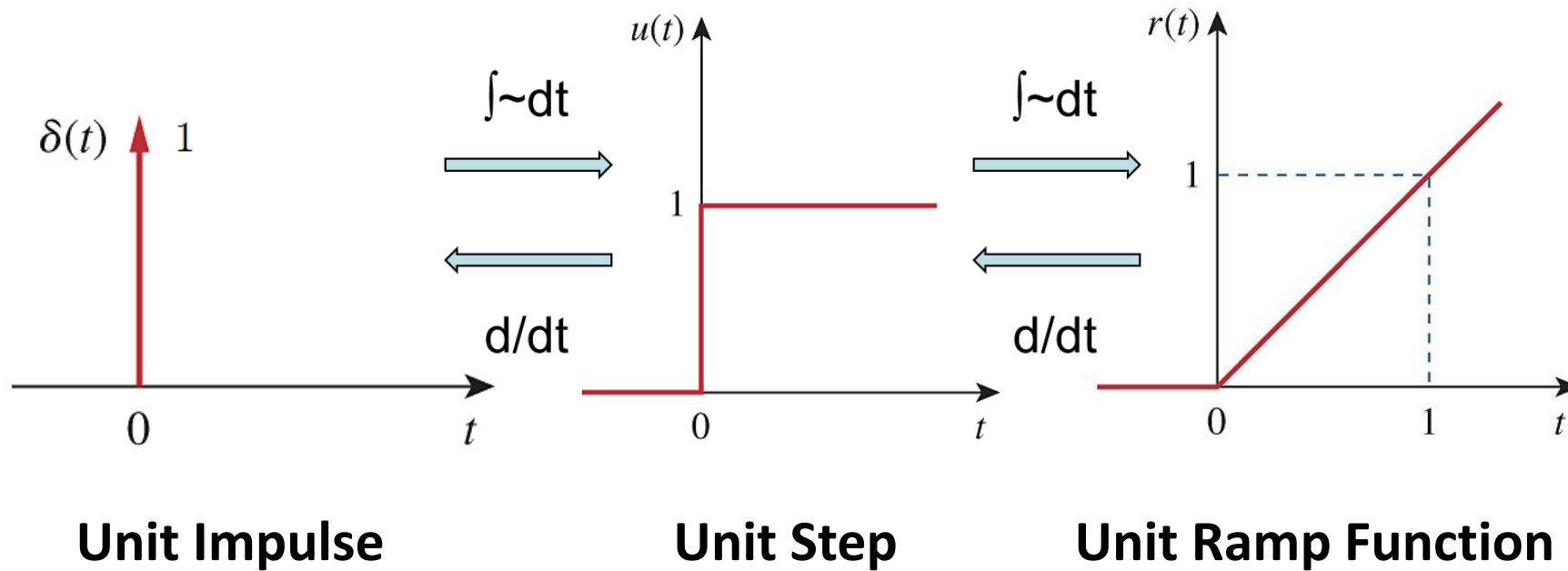
$$i(t) = i(0)e^{-t/\tau} = 2e^{-t/0.5} = 2e^{-2t} \text{ (A)}$$

7.4 Singularity Functions

- Singularity functions (also called switching functions) are functions that either are **discontinuous** or have **discontinuous derivatives**. They serve as good approximations to switching operations.



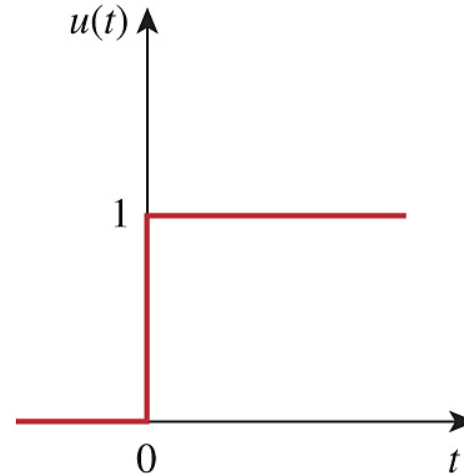
The three most widely used singularity functions in circuit analysis are the *unit step*, the *unit impulse*, and the *unit ramp* functions.



(i) Unit Step

The unit step function $u(t)$ is 0 for negative values of t and 1 for positive values of t . In mathematical terms,

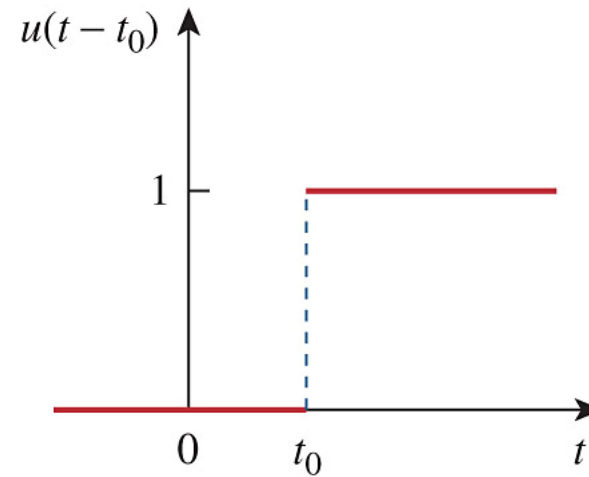
$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



The unit step function is not defined at $t = 0$, where it changes abruptly from 0 to 1.

If the abrupt change occurs at $t = t_0$ (where $t_0 > 0$) instead of $t = 0$, the mathematical representation becomes

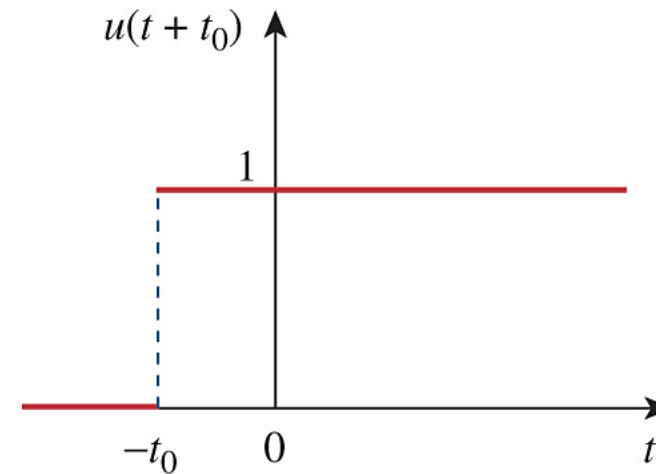
$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$



meaning that $u(t)$ is delayed by t_0 seconds.

If the abrupt change occurs at $t = -t_0$ (where $t_0 > 0$) instead of $t = 0$, the mathematical representation becomes

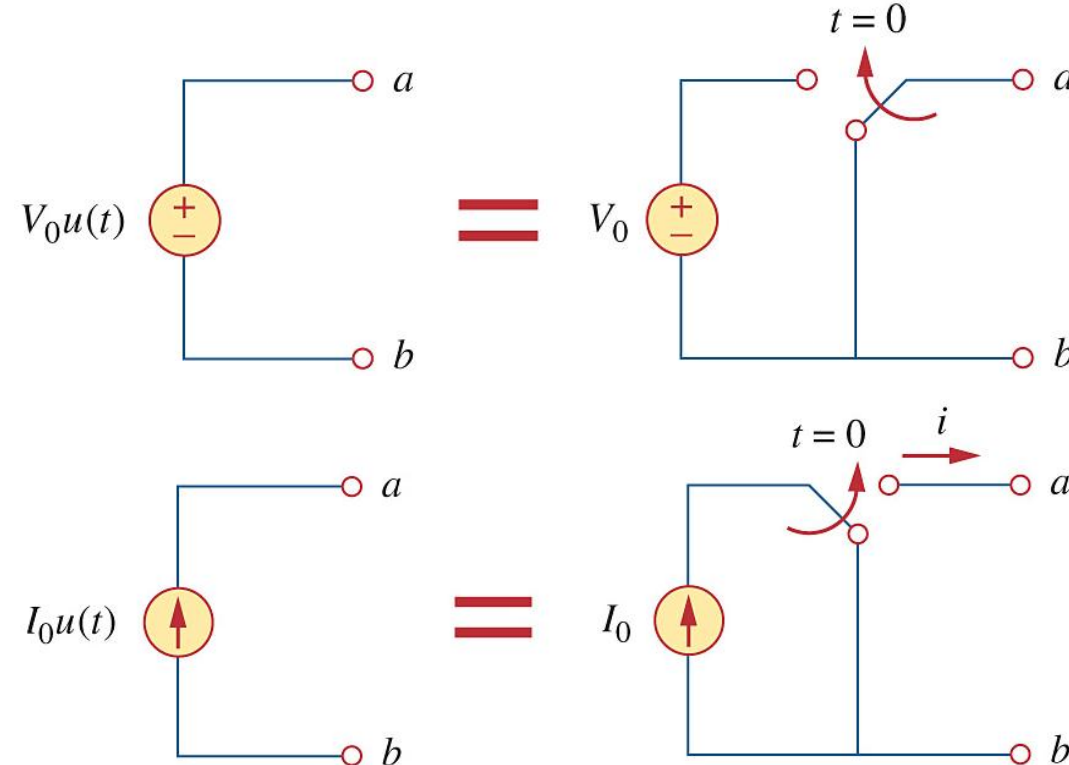
$$u(t + t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$



meaning that $u(t)$ is advanced by t_0 seconds.

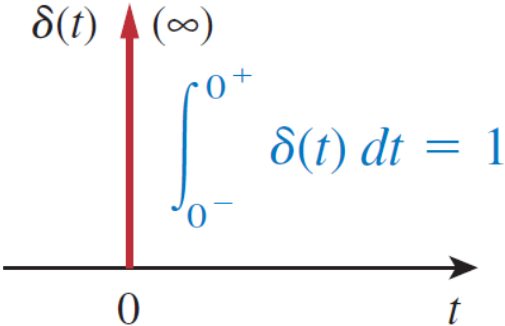
Switching operations

Switching operations create abrupt changes in voltages and currents. An abrupt change can be **represented by the step function**, $Ku(t)$, where K is a constant.



(ii) Unit Impulse

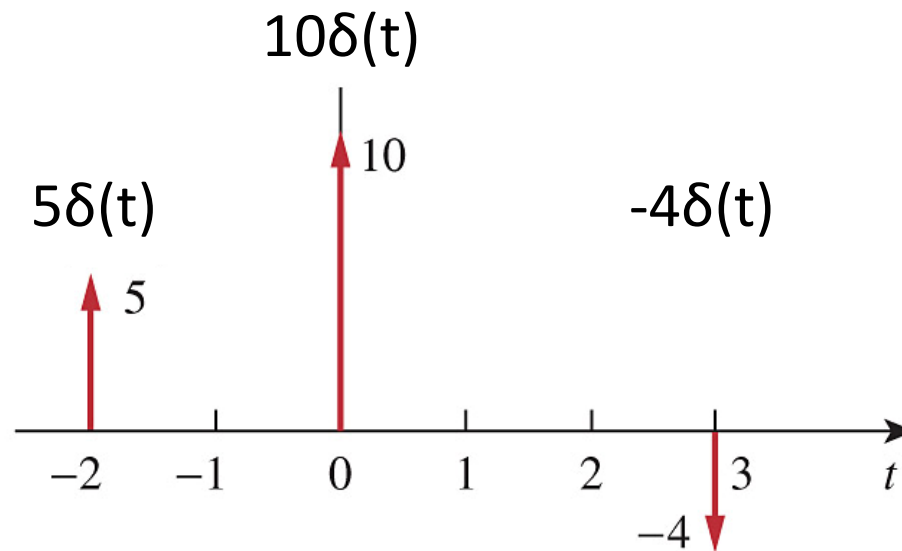
The derivative of the unit step function $u(t)$ is the unit impulse function $\delta(t)$, or delta function.

$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$
$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$


The **unit impulse function** $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined, i.e. a signal of infinite amplitude and zero duration.

The **area** is known as the **strength** of the function.

e.g. $10\delta(t)$ has an area of 10.



The unit impulse function has a property:

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$$

where $f(t)$ is a function that is continuous at $t = t_0$.

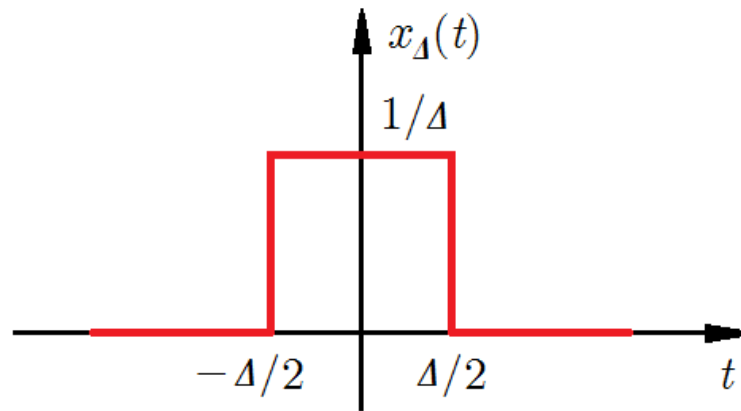
This means: when a function is integrated with the impulse function, we obtain the value of the function at the point where the impulse occurs → **sampling or sifting property**.

Proof :

$$\begin{aligned}\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt &= \int_{-\infty}^{\infty} f(t_0)\delta(t-t_0)dt \\ &= f(t_0)\int_{-\infty}^{\infty} \delta(t-t_0)dt = f(t_0)\end{aligned}$$

Unit Impulse Another Approach

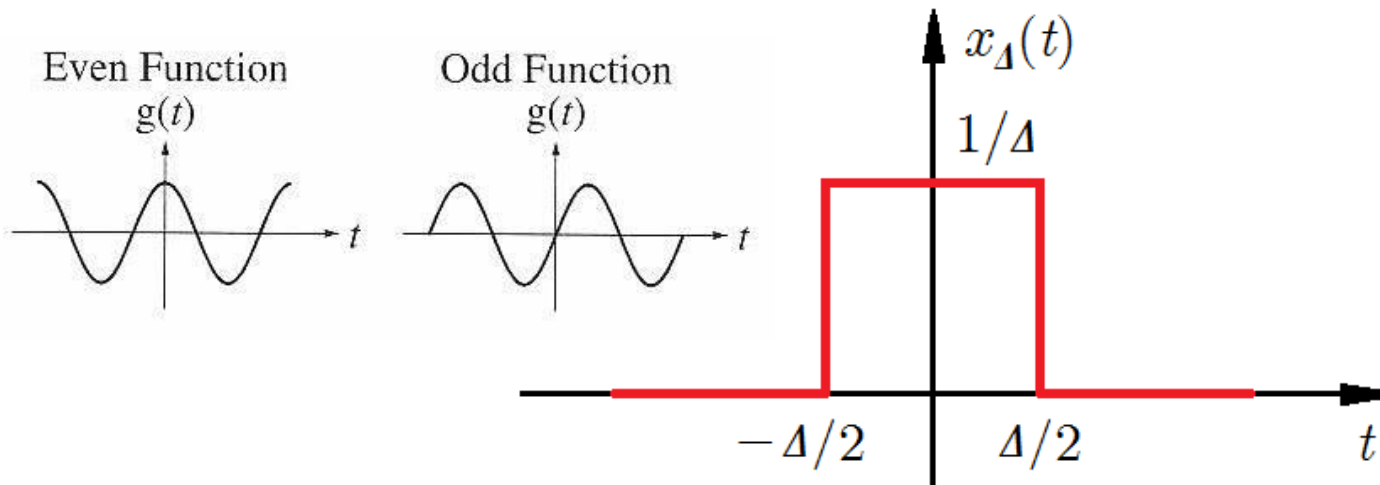
Singularity functions can be used to write mathematical expressions for signals. For example, the rectangular pulse



$$x_{\Delta}(t) = \frac{1}{\Delta} \left(u(t + \Delta / 2) - u(t - \Delta / 2) \right)$$

We view the unit impulse function $\delta(t)$ as the limiting form of any pulse, say the rectangular pulse $x_\Delta(t)$ (shown below), that is an even function of time t with unit area:

$$\delta(t) = \lim_{\Delta \rightarrow 0} x_\Delta(t)$$



Proof

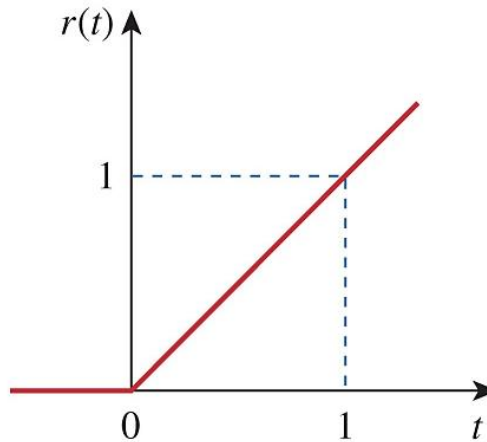
Without loss of generality, we take the rectangular pulse as an example.

$$\begin{aligned}\delta(t) &= \lim_{\Delta \rightarrow 0} x_{\Delta}(t) \\ &= \lim_{\Delta \rightarrow 0} \frac{1}{\Delta} \left(u(t + \Delta / 2) - u(t - \Delta / 2) \right) \\ &= \lim_{\Delta \rightarrow 0} \frac{u(t + \Delta / 2) - u(t - \Delta / 2)}{(t + \Delta / 2) - (t - \Delta / 2)} = \frac{d}{dt} u(t)\end{aligned}$$

(iii) Unit Ramp

The unit ramp function is zero for negative values of t and has a unit slope for positive values of t .

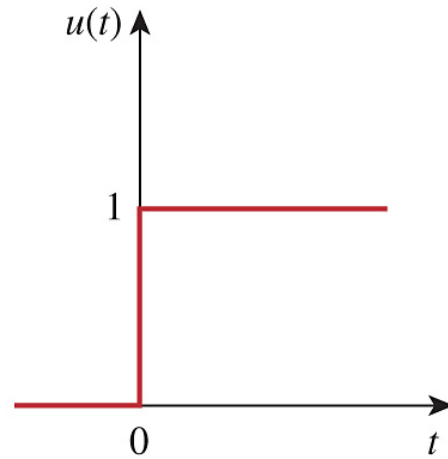
$$r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$



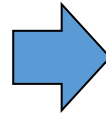
Integrating the unit step function $u(t)$ results in the unit ramp function $r(t)$.

$$r(t) = \int_{-\infty}^t u(\tau) d\tau = tu(t)$$

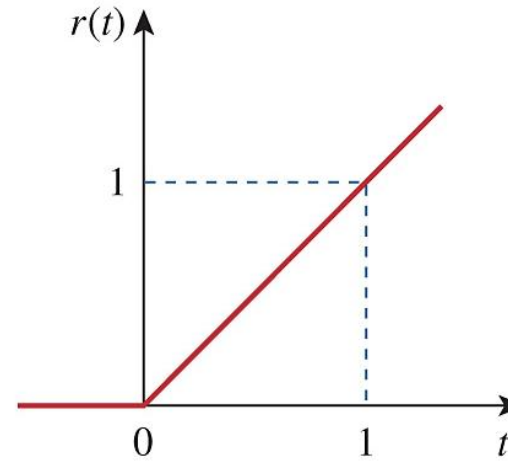
Unit step



$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



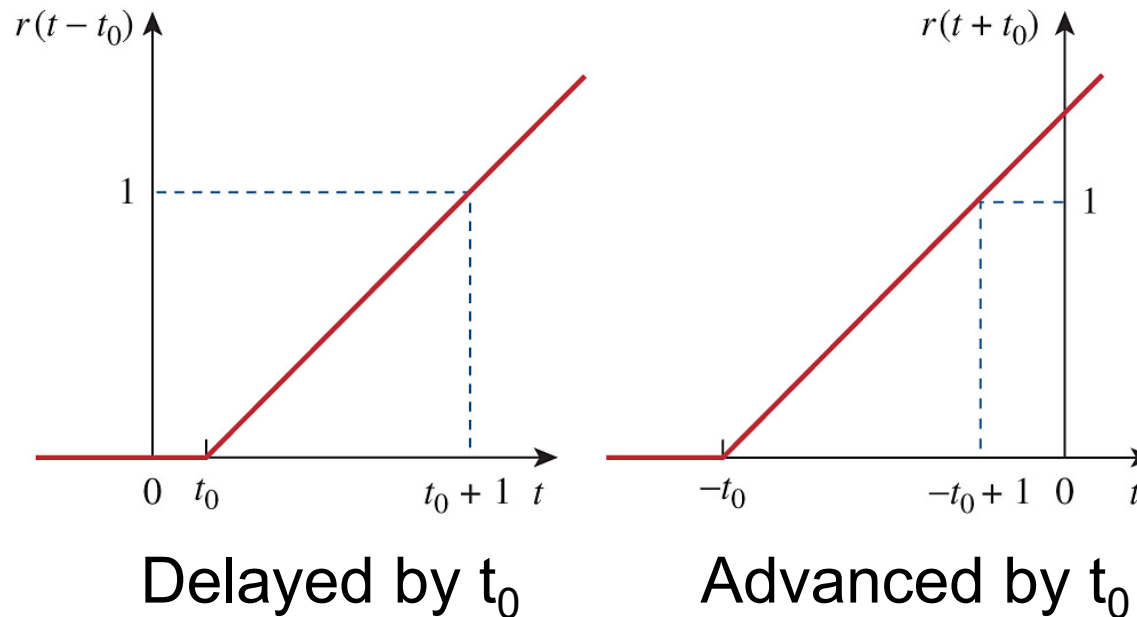
Unit ramp



$$r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$

Time delay in the unit ramp function

The unit ramp function may be delayed or advanced by time t_0 .



Example 7.6 Express the voltage pulse in Fig. 7.31 in terms of the unit step. Calculate its derivative and sketch it.

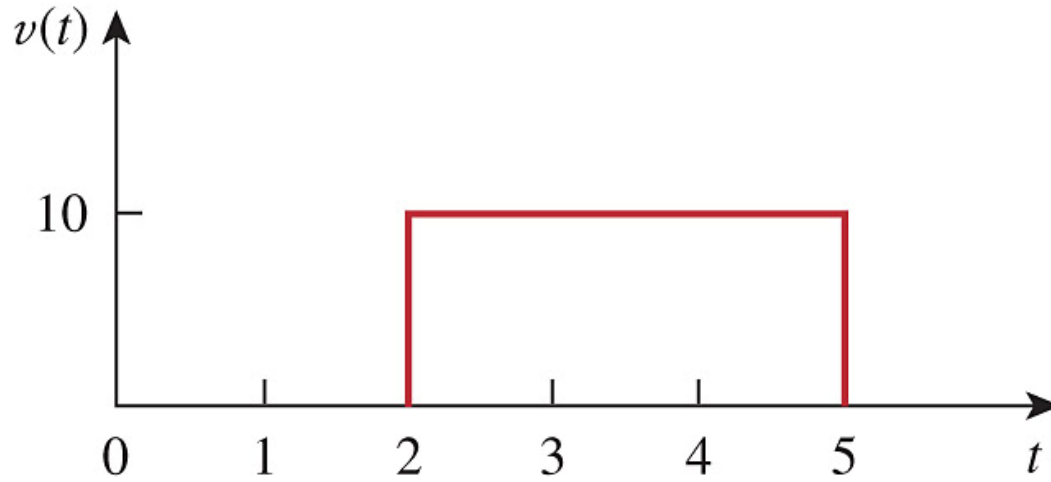


Figure 7.31

Solution :

$$v(t) = 10[u(t-2) - u(t-5)]$$

$$\frac{dv}{dt} = 10[\delta(t-2) - \delta(t-5)], \text{ see Fig. 7.32(b).}$$

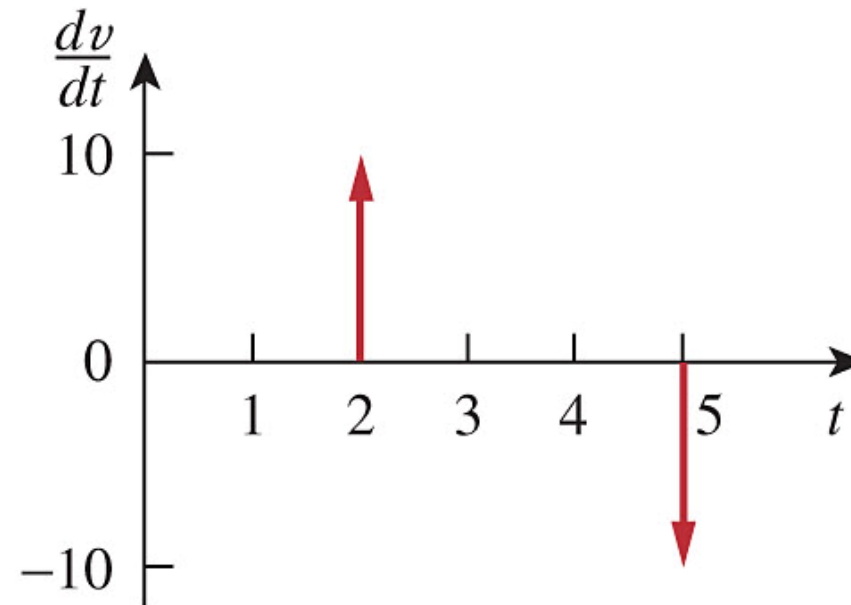


Figure 7.32(b)

Example 7.7 Express the *sawtooth* function shown in Fig. 7.35 in terms of singularity functions.

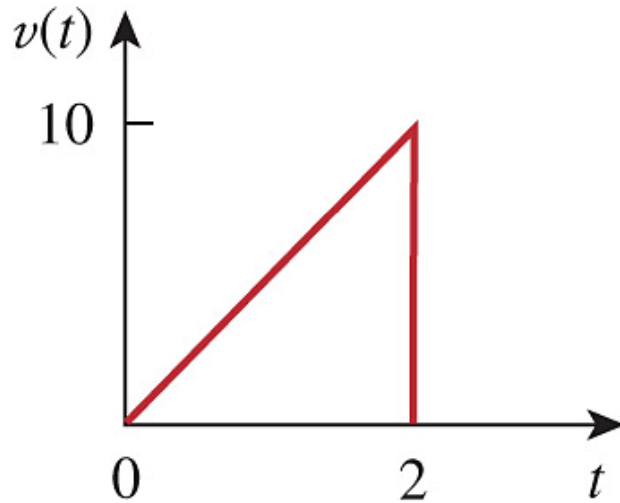
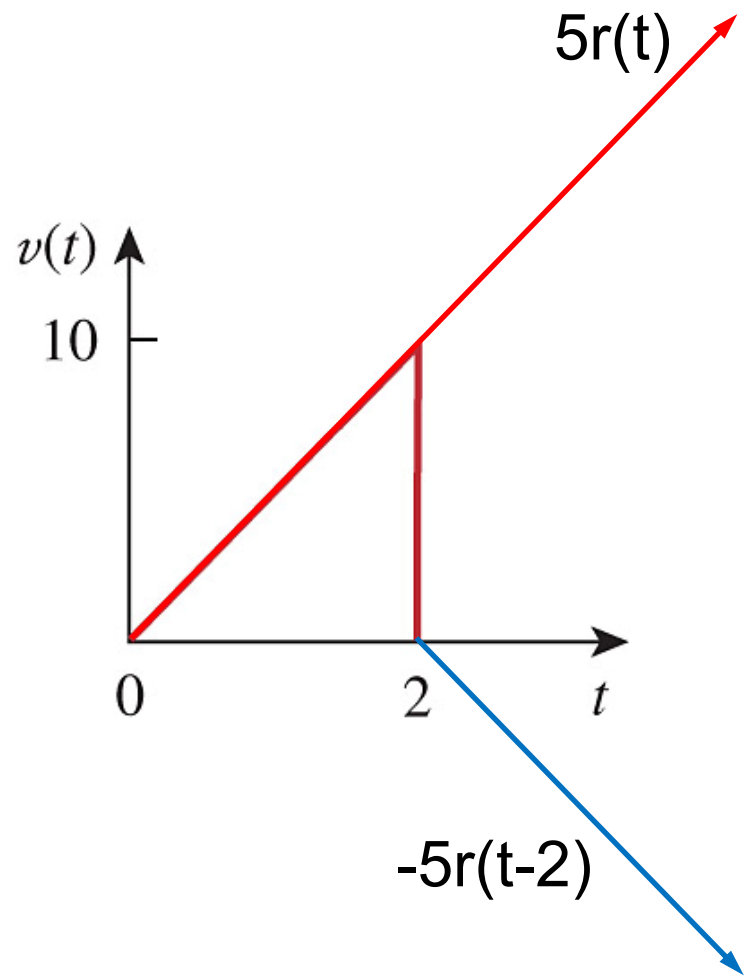
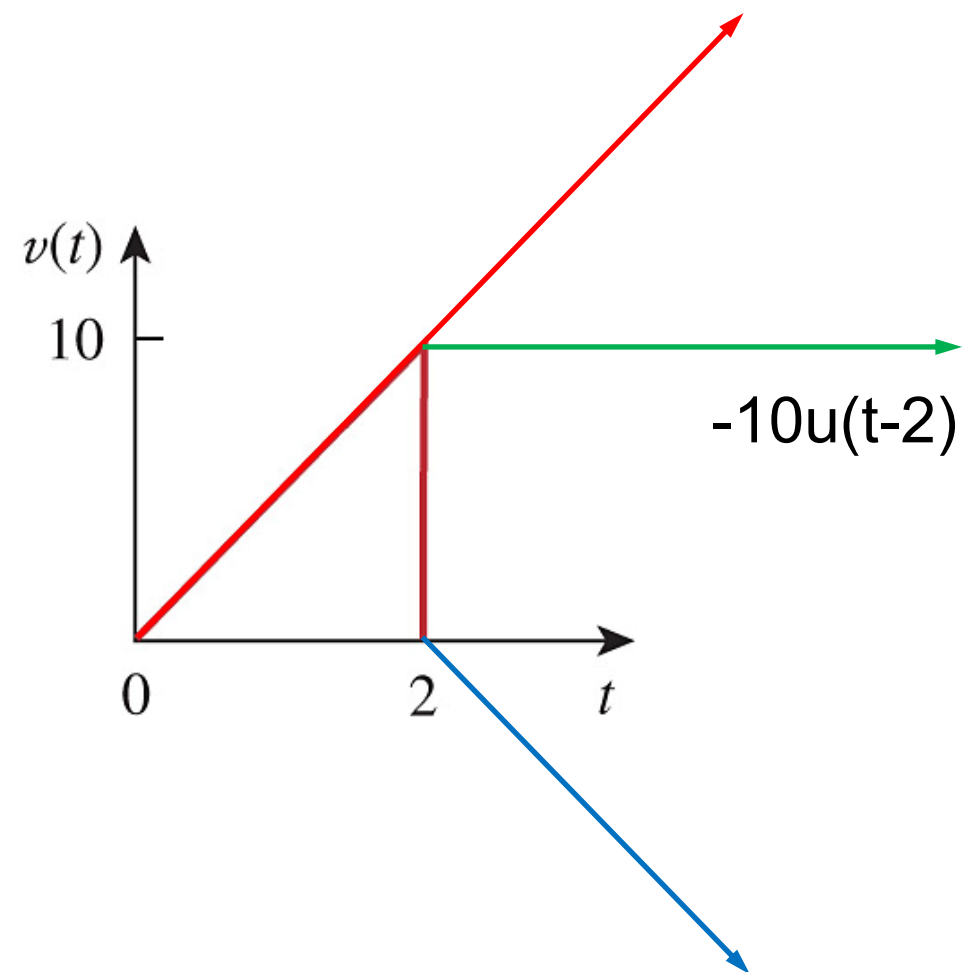
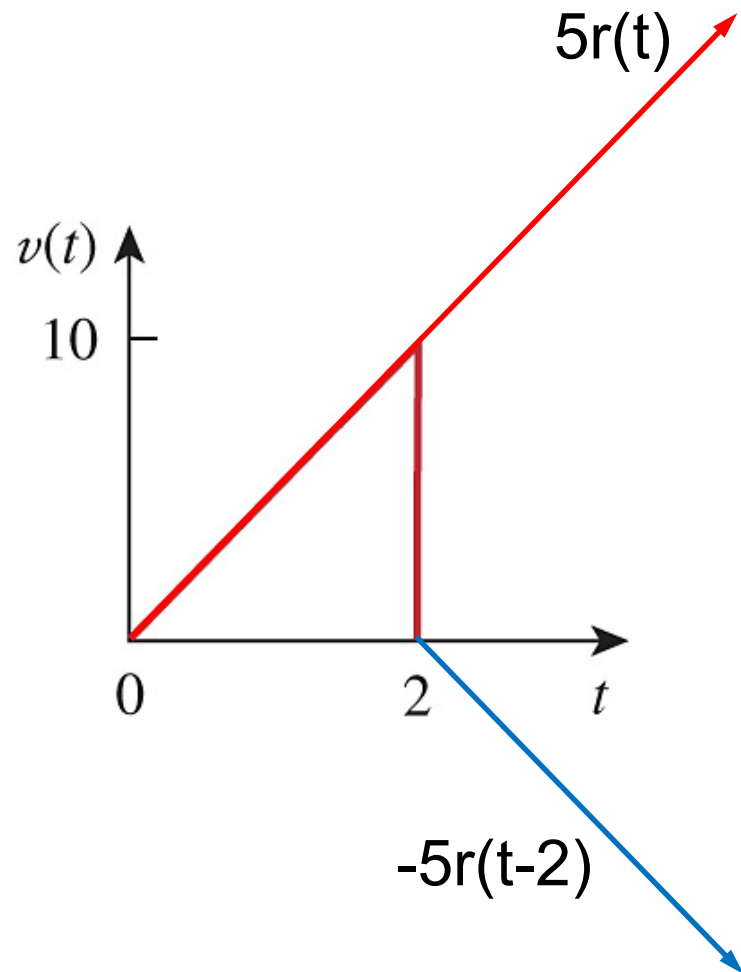


Figure 7.35

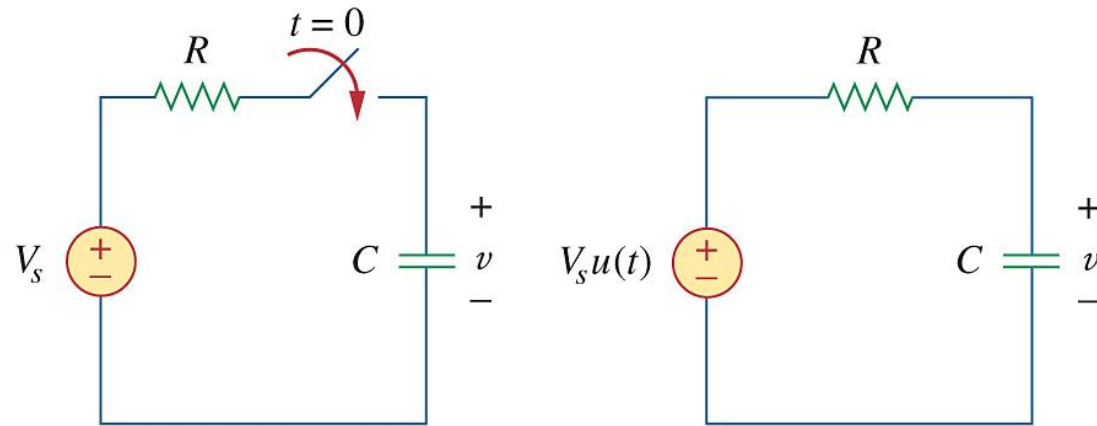




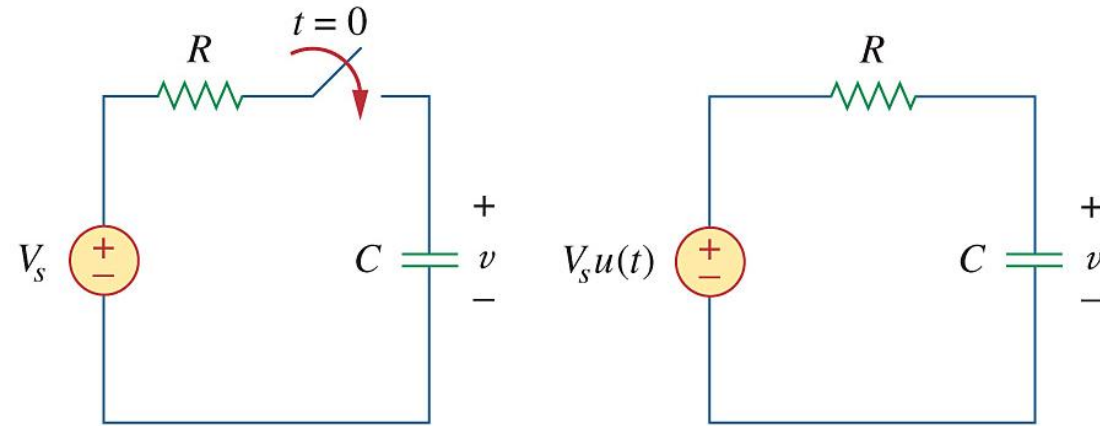
$$v(t) = 5r(t) - 5r(t-2) - 10u(t-2)$$

7.5 Step Response of an RC Circuit

The **step response** of a circuit is its behavior due to a sudden application of **the step function**, either a voltage or a current source.

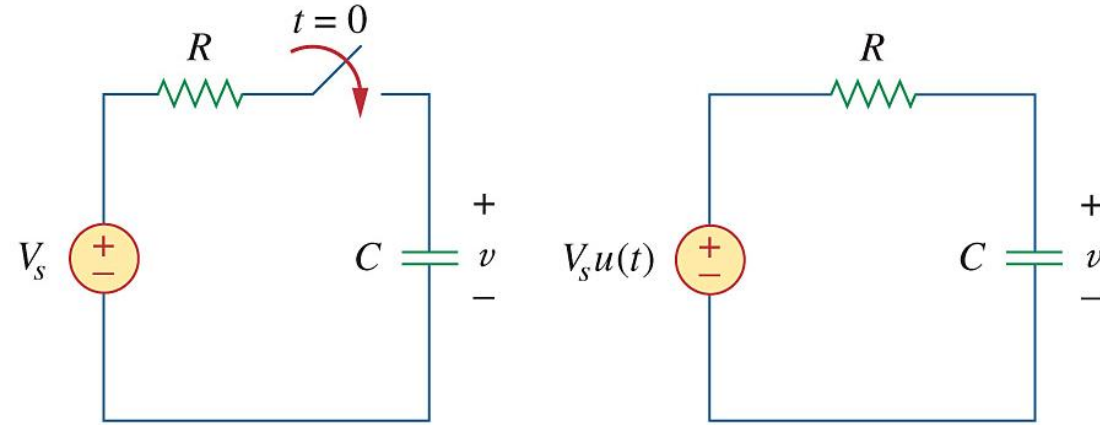


We select **the capacitor voltage as the circuit response** to be determined. We assume an initial voltage V_0 on the capacitor.



Because the voltage across the capacitor cannot change instantaneously

$$\boxed{v(0^+) = v(0^-) = V_0} \quad \text{Initial condition}$$



For $t > 0$,

$$\left(C \frac{dv}{dt} \right) R + v = V_s \Rightarrow \frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

or

$$\frac{dv}{dt} + \frac{1}{\tau} v = \frac{V_s}{\tau}, \text{ where } \tau = RC$$

Solve this differential equation

Method 1

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC} \quad \Rightarrow \quad \frac{dv}{v - V_s} = -\frac{dt}{RC} \quad \text{Integrate both sides}$$

$$\ln(v - V_s) \Big|_{V_0}^{v(t)} = -\frac{t}{RC} \Big|_0^t \quad \Rightarrow \quad \ln(v(t) - V_s) - \ln(V_0 - V_s) = -\frac{t}{RC} + 0$$

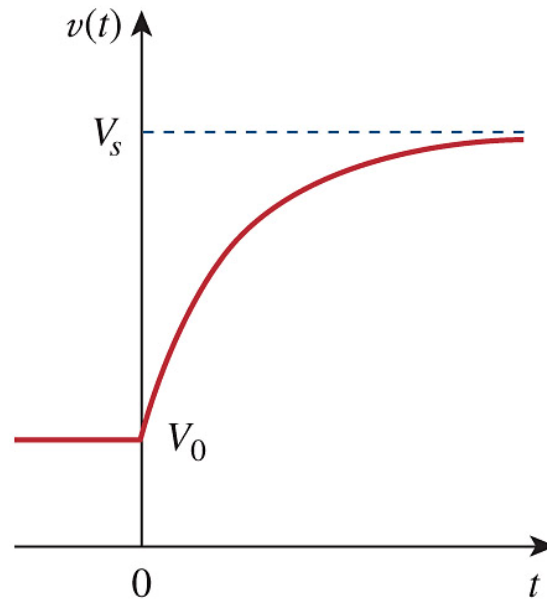
$$\ln\left(\frac{v - V_s}{V_0 - V_s}\right) = -\frac{t}{RC} \quad \Rightarrow \quad \frac{v - V_s}{V_0 - V_s} = e^{-t/\tau}, \quad \tau = RC$$

Finally we get $v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$

Thus, the complete response (or total response) is

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$

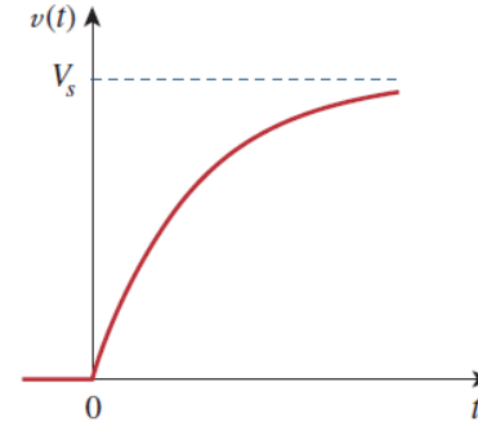
Assuming that $V_s > V_0$ a plot of $v(t)$ is



If we assume that the capacitor is **uncharged initially**

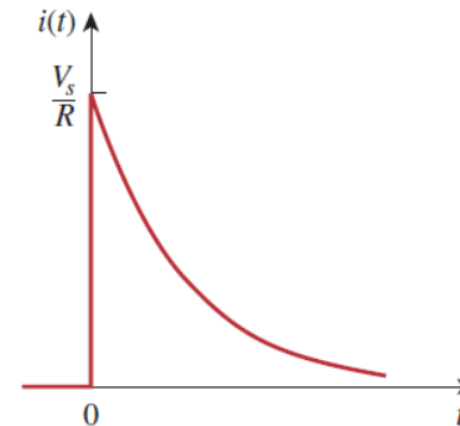
$$v(t) = \begin{cases} 0, & t < 0 \\ V_s(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

Or $v(t) = V_s(1 - e^{-t/\tau})u(t)$



The current through the capacitor is $i(t) = C \frac{dv}{dt}$ thus we get

$$i(t) = \frac{V_s}{R} e^{-t/\tau} u(t)$$



Method 2

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC} \quad \text{use a test solution } v(t) = Ae^{rt}$$

$$r + \frac{1}{\tau} = 0 \Rightarrow r = -\frac{1}{\tau}$$

(i) The homogeneous solution or *natural response*

$$v_n(t) = Ae^{-t/\tau}$$

(ii) Suppose the particular solution or force response

$$v_f(t) = B$$

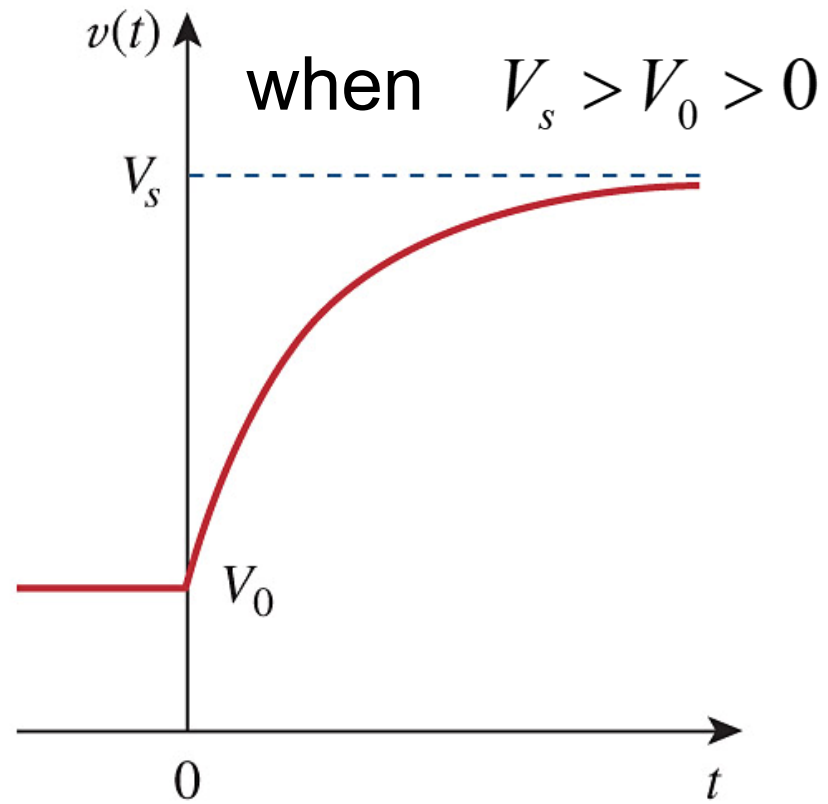
$$\frac{dB}{dt} + \frac{1}{\tau}B = \frac{V_s}{\tau} \Rightarrow B = V_s$$

The complete solution (= complete response or total response)

$$\left. \begin{aligned} v(t) &= v_n(t) + v_f(t) = Ae^{-t/\tau} + V_s \\ \text{When } t &= 0^+, \\ v(0^+) &= A + V_s = V_0 \Rightarrow A = V_0 - V_s \end{aligned} \right\} v(t) = (V_0 - V_s)e^{-t/\tau} + V_s$$

This is **the response of the RC circuit** to a sudden application of a constant dc source, assuming the capacitor is initially charged.

$$v(t) = (V_0 - V_s)e^{-t/\tau} + V_s$$



If the capacitor is **uncharged** initially, i.e., $V_0 = 0$, then

$$v(t) = V_s(1 - e^{t/\tau}), \quad t > 0 \quad \text{or} \quad v(t) = V_s(1 - e^{t/\tau})u(t)$$

This is the **zero-state response**. The zero-state response corresponding to a unit-step input is called the unit-step response.

Interpretations of the response

From the equation $v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$

We can have two interpretations:

(i) Complete response = natural (v_n) + forced response(v_f)

$$v_n = V_0 e^{-t/\tau} \quad v_f = V_s (1 - e^{-t/\tau})$$

(ii) Complete response = transient (v_t) + steady-state (v_{ss})

$$v_t = (V_0 - V_s)e^{-t/\tau} \quad v_{ss} = V_s$$

$$v(t) = V_S + (V_0 - V_S)e^{-t/\tau}, \quad t > 0$$

(i) Complete response = natural (v_n) + forced response(v_f)

$$v_n = V_0 e^{-t/\tau} \quad v_f = V_S(1 - e^{-t/\tau})$$

The natural response (v_n) is from the energy initially stored.

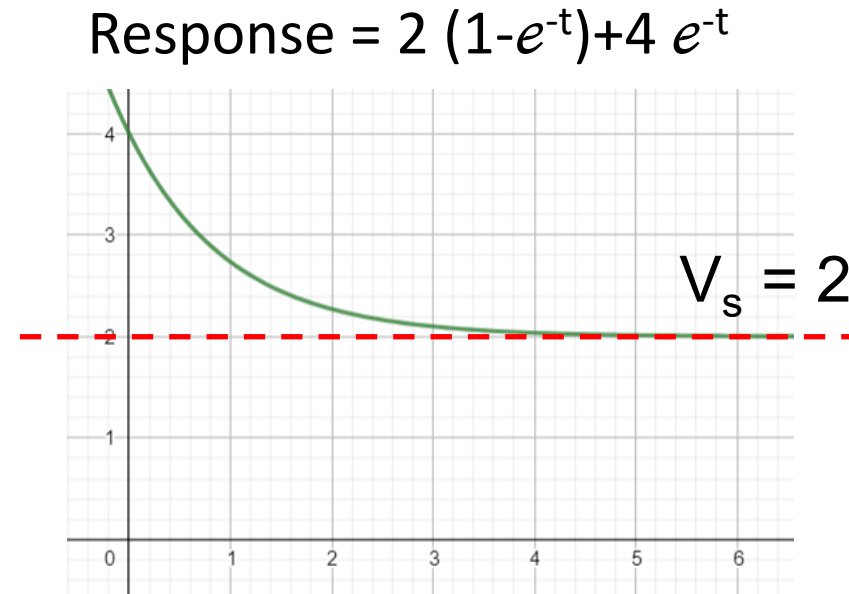
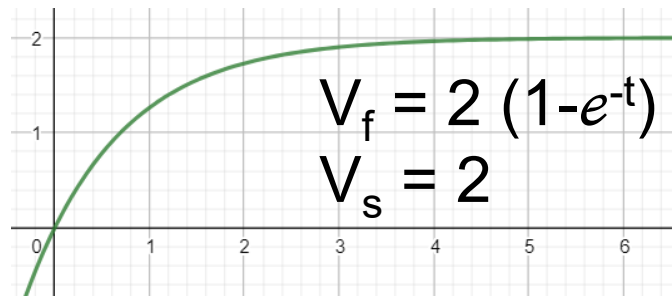
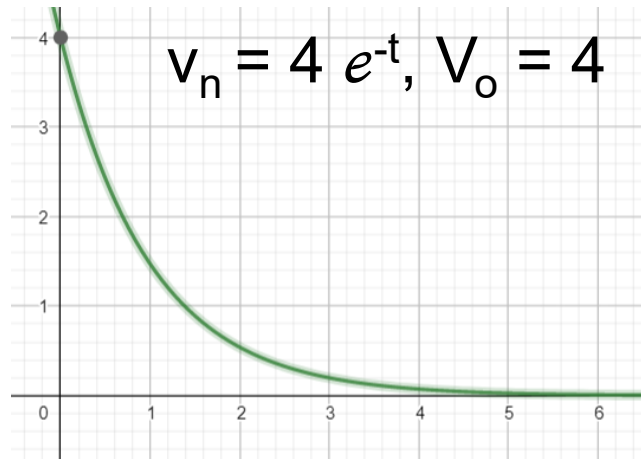
The forced response (v_f) is by an external force (power sources).

Exponential signals die out eventually, **leaving only the steady-state component** of the forced response.

(i) Complete response = natural (v_n) + forced response(v_f)

$$v_n = V_o e^{-t/\tau} \quad v_f = V_s(1 - e^{-t/\tau})$$

$$V_o > V_s > 0$$



$$v(t) = V_S + (V_0 - V_S)e^{-t/\tau}, \quad t > 0$$

(ii) Complete response = transient (v_t) + steady-state (v_{ss})

$$v_t = (V_0 - V_S)e^{-t/\tau} \quad v_{ss} = V_S$$

The transient response is the temporary response that will die out with time.

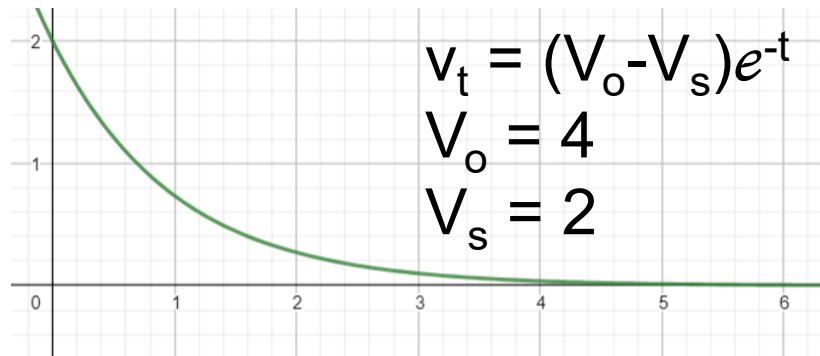
The steady-state response is the behavior of the circuit a long time after an external excitation is applied.

Under certain conditions, v_n and v_t response are the same. The same can be said about the v_f and v_{ss} response.

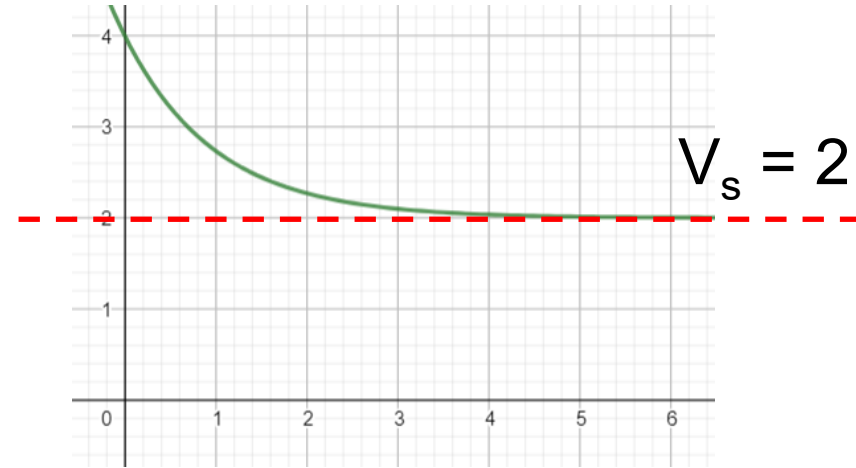
(ii) Complete response = transient (v_t) + steady-state (v_{ss})

$$v_t = (V_o - V_s)e^{-t/\tau} \quad v_{ss} = V_s$$

$V_o > V_s > 0$



Response = $V_s + (V_o - V_s)e^{-t}$



The complete response $v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$

may be written as $v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$

$v(0)$ is the initial voltage at $t = 0^+$

$v(\infty)$ is the final or steady-state value

Thus, to find the step response of an RC circuit requires **three parameters**:

1. The initial capacitor voltage $v(0)$
2. The final capacitor voltage $v(\infty)$
3. The time constant τ .

Example 7.10 The switch in Fig.7.43 has been in position A for a long time. At $t = 0$, the switch moves to B . Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4s.

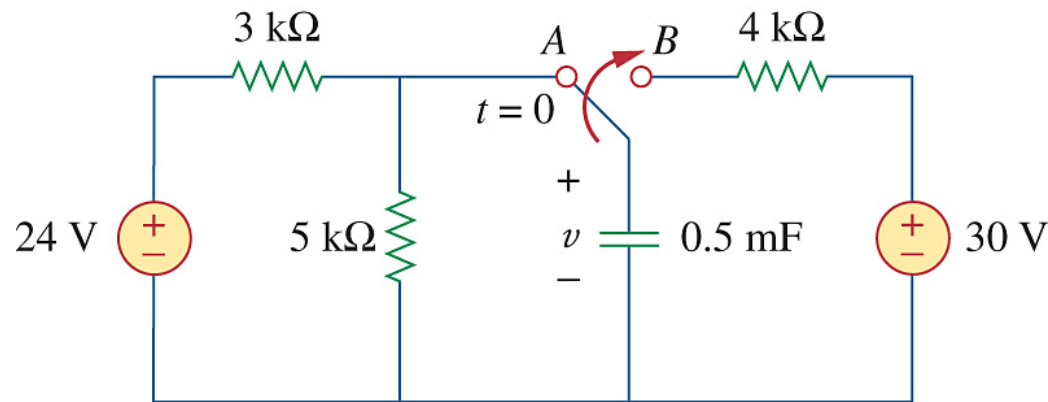


Figure 7.43

Example 7.10 The switch in Fig.7.43 has been in position A for a long time. At $t = 0$, the switch moves to B . Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4s.

Solution :

For $t < 0$,

$$v(t) = 24 \times \frac{5}{3+5} = 15 \text{ (V)}$$

$$v(0^-) = 15 \text{ V}$$

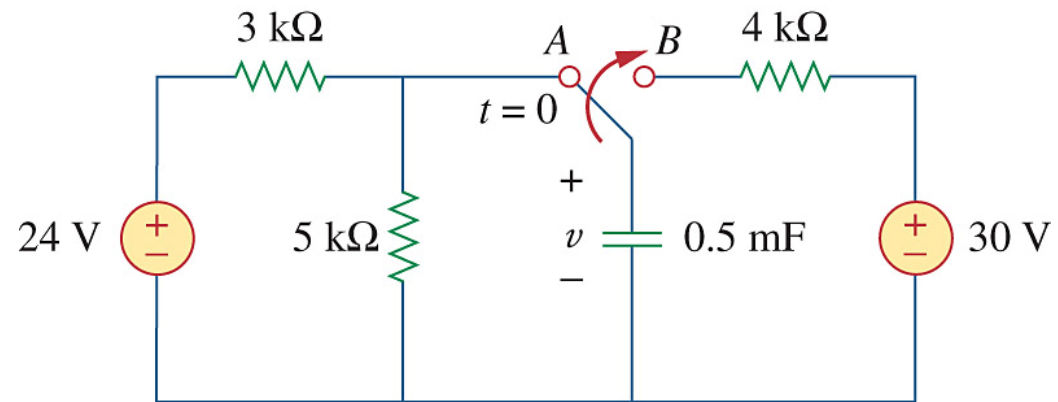


Figure 7.43

For $t > 0$,

$$v(0^+) = v(0^-) = 15 \text{ V}$$

$$v(\infty) = 30 \text{ V}$$

$$\tau = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ (s)}$$

$$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau}$$

$$= 30 + (15 - 30)e^{-t/2} = 30 - 15e^{-0.5t} \text{ (V)}$$

$$v(1) = 30 - 15e^{-0.5 \times 1} \approx 20.90 \text{ (V)}$$

$$v(4) = 30 - 15e^{-0.5 \times 4} \approx 27.97 \text{ (V)}$$

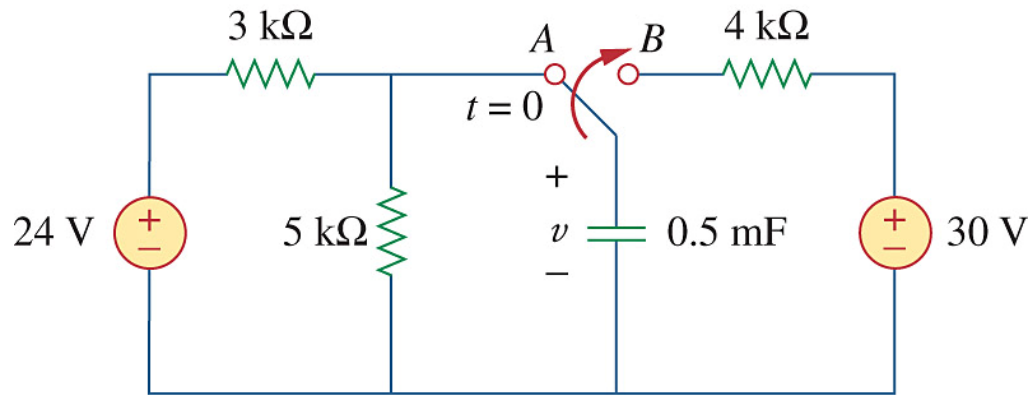


Figure 7.43

1st Order ODE (Optional)

$$y' + p(x)y = r(x) \rightarrow \mathbf{F}y' + \mathbf{F}p(x)\mathbf{y} = \mathbf{F}r(x)$$

$$\text{Left side: } (Fy)' = Fy' + F'y$$

$$\text{where } F'y = Fp(x)y, \text{ then } \mathbf{F}' = \mathbf{F}p(x)$$

$$F' = Fp(x) \rightarrow \frac{1}{F}F' = p(x) \rightarrow \ln F = \int p(x)dx$$

$$\text{Thus, } \mathbf{F} = \mathbf{e}^{\int p(x)dx} \text{ where we set } \int \mathbf{p}(x)d\mathbf{x} = \mathbf{h}$$

$$\mathbf{F} = \mathbf{e}^{\mathbf{h}} \text{ then, } \mathbf{p} = \mathbf{h}' \text{ due to } h = \int p(x)dx$$

$$Fy' + Fp(x)y = Fr(x) \rightarrow e^h y' + e^h p(x)y = e^h r(x) \\ \rightarrow \mathbf{e^h y' + e^h h' y = e^h r(x)}$$

$$(e^h y)' = e^h r(x) \text{ integral both sides}$$

$$e^h y = \int e^h r(x) dx + c$$

$$\mathbf{y = e^{-h}(\int e^h r(x) dx + c) \text{ where } h = \int p(x) dx}$$

1st Order ODE where $r(x) = 0$

$$y' + p(x)y = r(x), \text{ where } r(x) = 0$$

$$\frac{dy}{dx} + p(x)y = 0$$

$$\rightarrow \frac{1}{y} dy = -p(x) dx \text{ integrate both sides}$$

$$\ln y = - \int p(x) dx + c \rightarrow \mathbf{y = c \cdot e^{- \int p(x) dx}}$$

$$\text{e.g. } \frac{di}{dt} + 5i = 0 \rightarrow i(t) = c \cdot e^{-5t}$$

Practice Problem 7.1 Refer to the circuit in Fig. 7.7. Let $v_C(0) = 45$ V. Determine v_C , v_x , and i_o for $t \geq 0$.

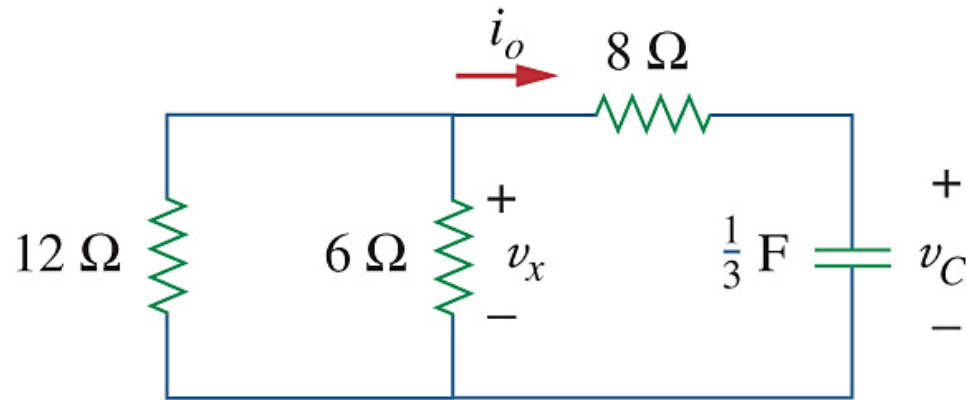
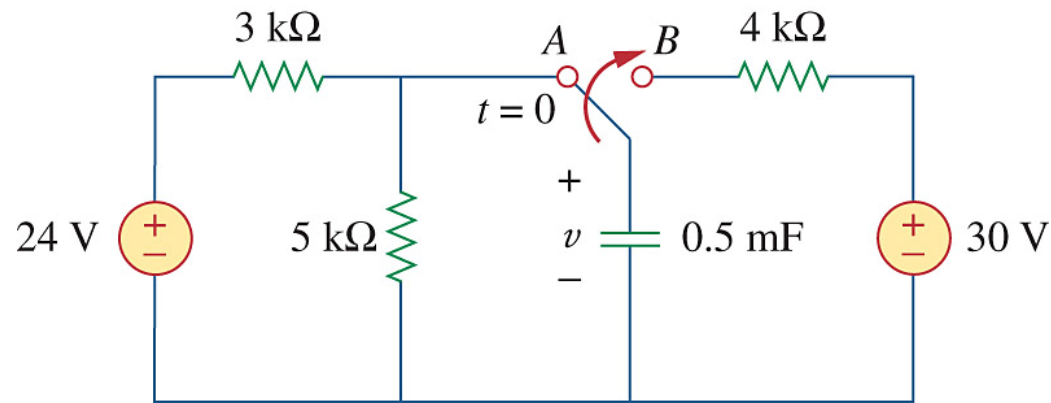


Figure 7.7 An RC circuit.

$$v_C = v_C(0)e^{-t/\tau} = 45e^{-t/4} = 45e^{-0.25t} \text{ (V)}$$

Example 7.10 The switch in Fig.7.43 has been in position *A* for a long time. At $t = 0$, the switch moves to *B*. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4s.



For $t < 0$,

$$v(t) = 24 \times \frac{5}{3+5} = 15 \text{ (V)}$$

$$v(0^-) = 15 \text{ V}$$

Practice Problem 7.11 The switch in Fig. 7.43 is closed at $t = 0$. Find $i(t)$ and $v(t)$ for all time.

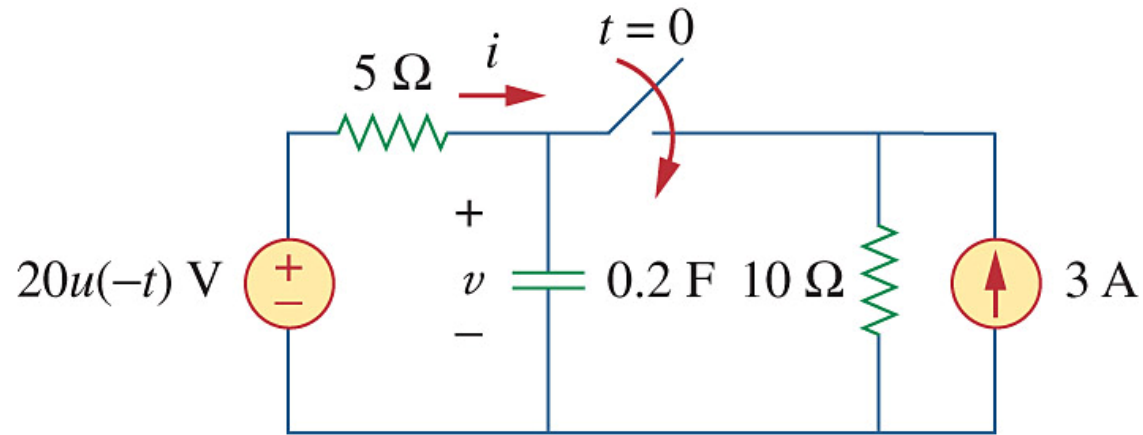


Figure 7.47

For $t > 0$,

$$v(0^+) = v(0^-) = 20 \text{ V}$$

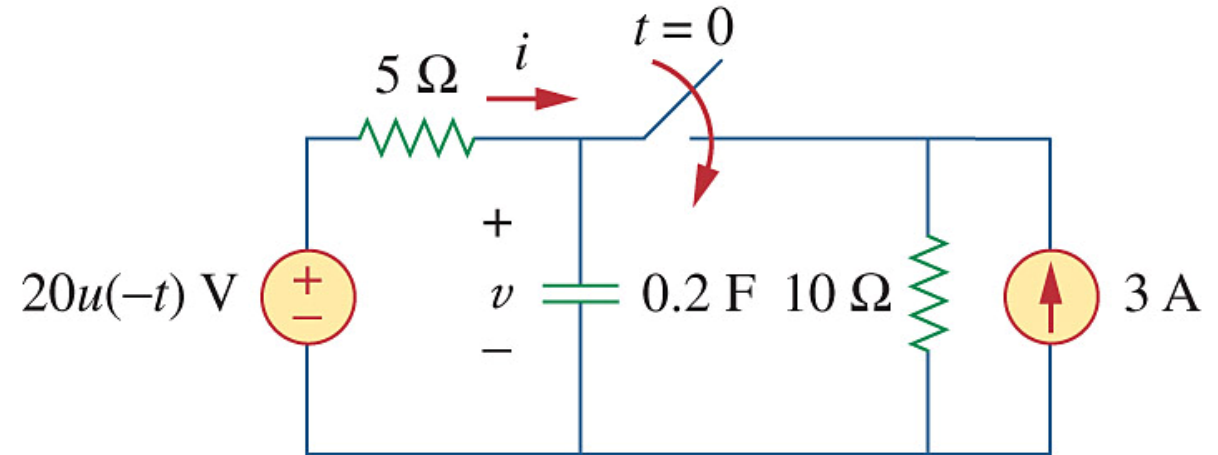
$$v(\infty) = 3 \times (5 \parallel 10) = 10 \text{ (V)}$$

$$\tau = (5 \parallel 10) \times 0.2 = \frac{2}{3} \text{ (s)}$$

$$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau}$$

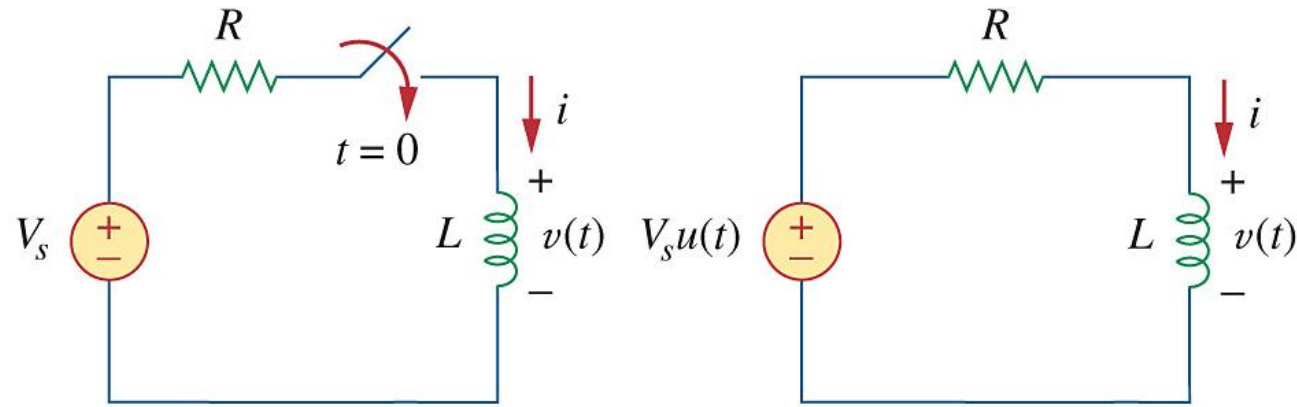
$$= 10 + (20 - 10)e^{-t/(2/3)} = 10(1 + e^{-1.5t}) \text{ (V)}$$

$$i(t) = -\frac{v(t)}{5} = -2(1 + e^{-1.5t}) \text{ (A)}$$

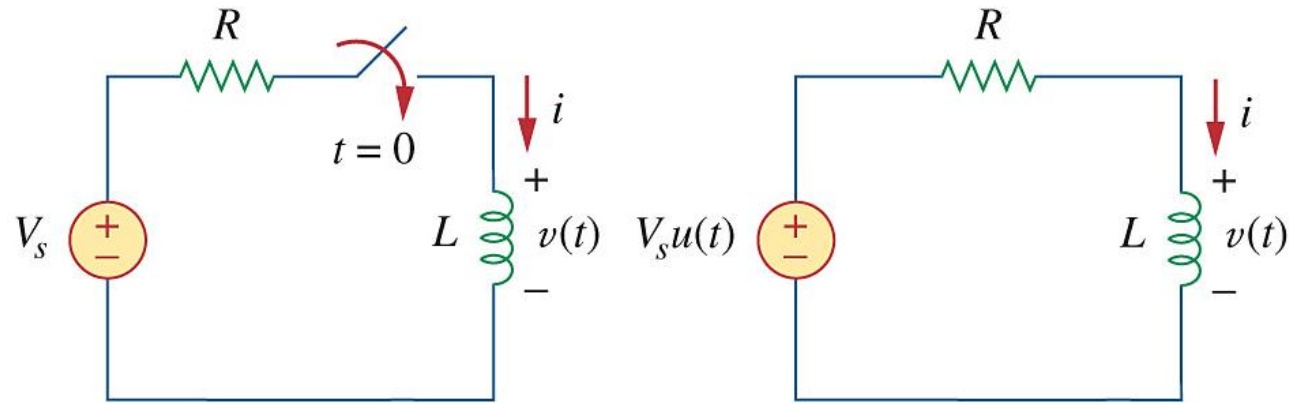


$v(0^+)$: by the value of $v(0^-)$
 $v(\infty)$: by the circuit at $t > 0$

7.6 Step Response of an RL Circuit



We select the inductor current i as the circuit response to be determined. We assume an initial current I_0 in the inductor.



Because the current through the inductor cannot change instantaneously, the initial condition is $i(0^+) = i(0^-) = I_0$

For $t > 0$,

$$iR + L \frac{di}{dt} = V_s$$

or

$$\frac{di}{dt} + \frac{1}{\tau} i = \frac{V_s / R}{\tau}, \text{ where } \tau = \frac{L}{R}$$

RL

$$\frac{di}{dt} + \frac{1}{\tau} i = \frac{V_s / R}{\tau}, \text{ where } \tau = \frac{L}{R}$$

RC

$$\frac{dv}{dt} + \frac{1}{\tau} v = \frac{V_s}{\tau}, \text{ where } \tau = RC$$

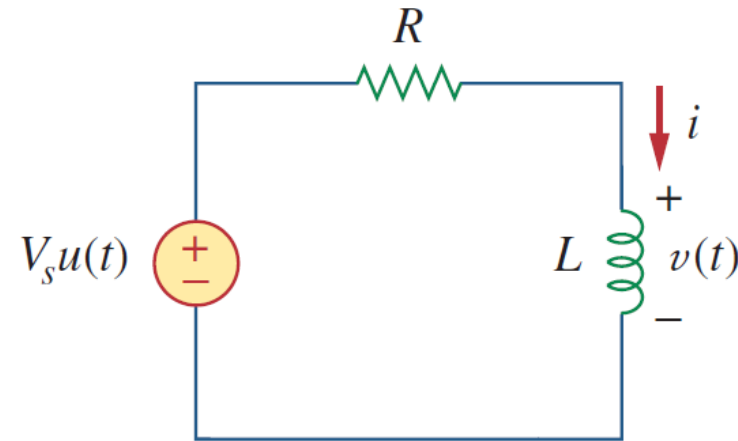
$$v(t) = (V_0 - V_s)e^{-t/\tau} + V_s$$

Since this differential equation has the same form as that describing the RC circuit,

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$

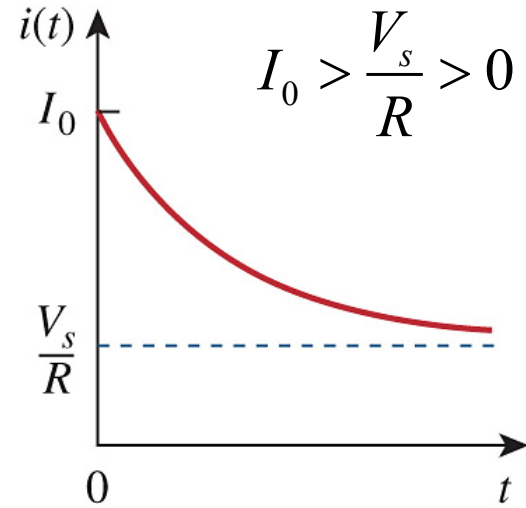
It is evident that

$$i(\infty) = \frac{V_s}{R}$$

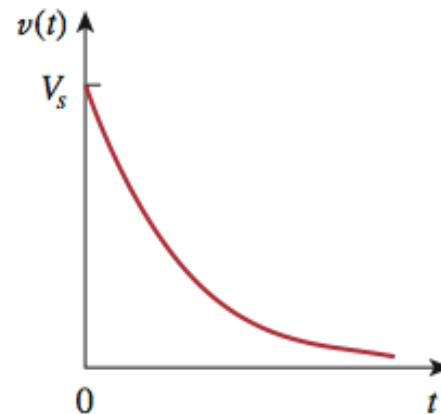
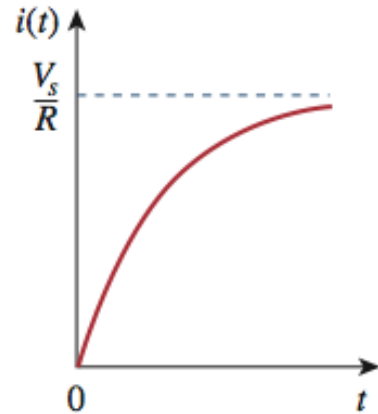


The plot of the RL step response is thus

$$i(t) = \frac{V_s}{R} + \left(I_0 - \frac{V_s}{R} \right) e^{-t/\tau}$$



If $I_0 = 0$, $i(t) = \frac{V_s}{R} \left(1 - e^{-t/\tau} \right) u(t)$, $v(t) = V_s e^{-t/\tau} u(t)$



The complete response can be written as

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty) \right] e^{-t/\tau}$$

Thus, to find the response of a first-order RL circuit requires three conditions:

1. The initial inductor current $i(0^+)$.
2. The final inductor current $i(\infty)$.
3. The time constant $\tau = L / R$.

Source-free RL circuit is a special case when $i(\infty) = 0$

Example 7.13 At $t = 0$, switch 1 in Fig. 7.53 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for $t > 0$. Calculate i for $t = 2$ s and $t = 5$ s.

Solution :

For $t < 0$,

$$i(t) = 0$$

$$i(0^-) = 0$$

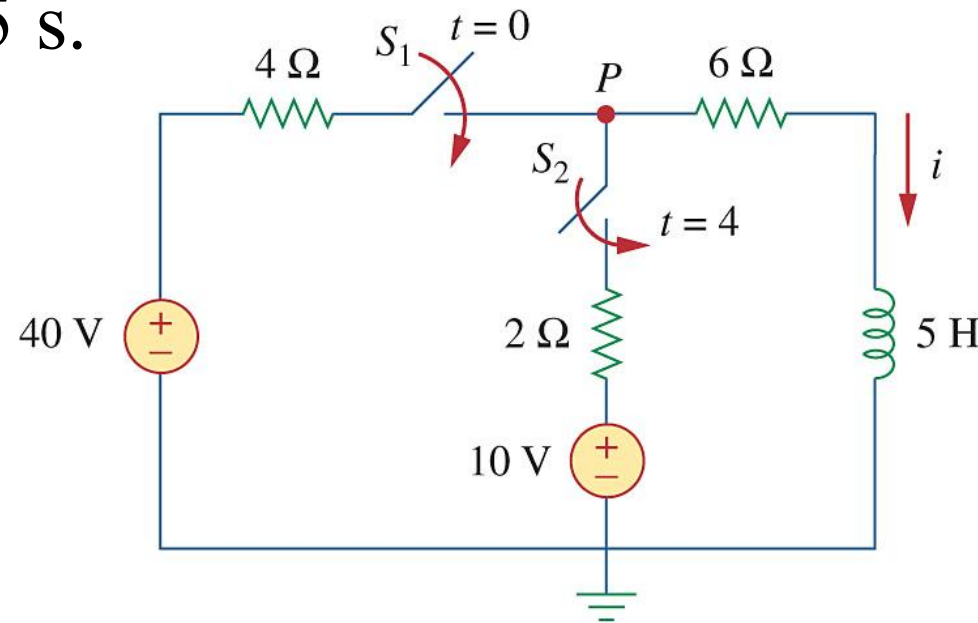


Figure 7.53

ii) $0 < t < 4$

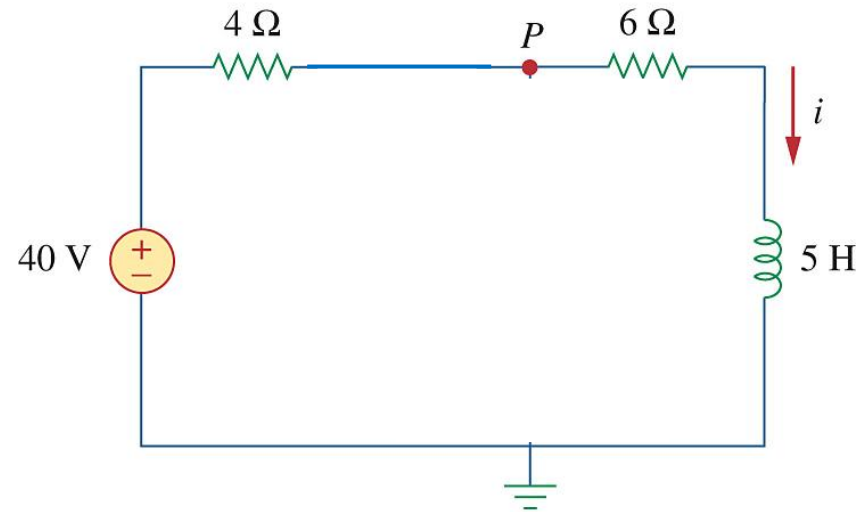
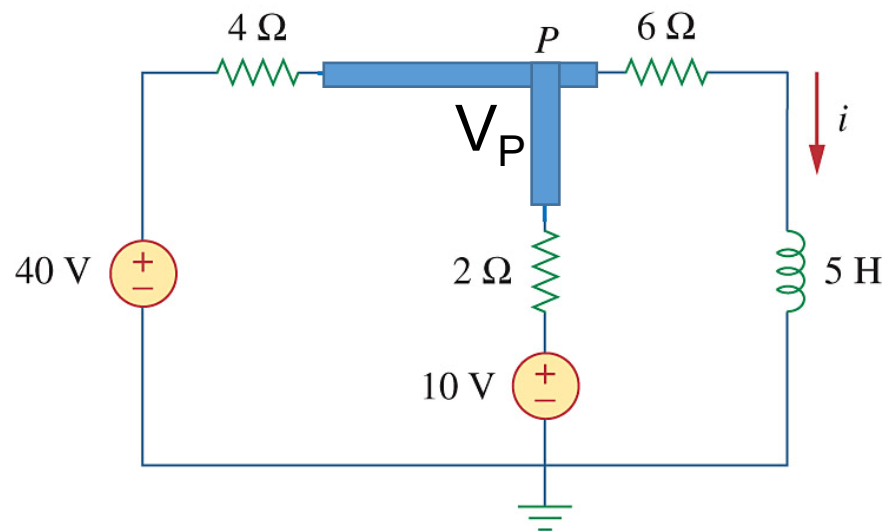
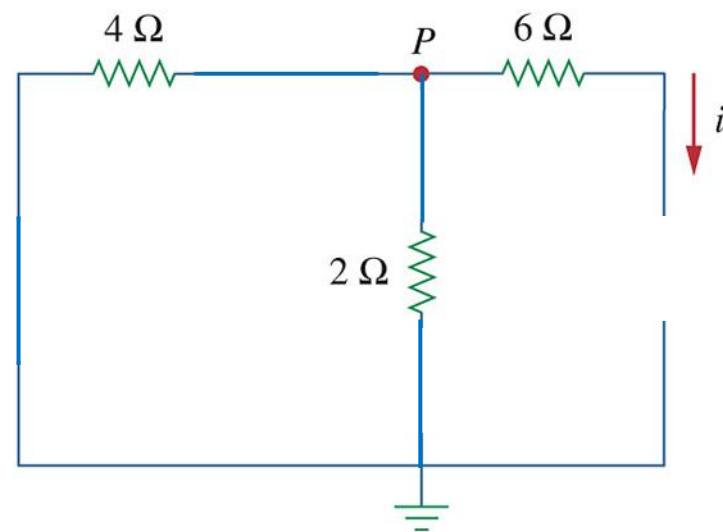


Figure 7.53

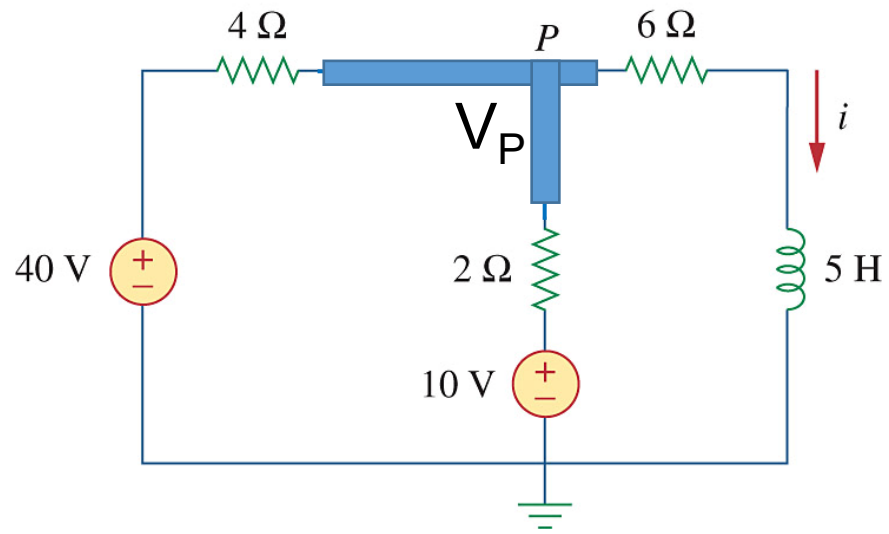
iii) $t > 4$



$$R_{eq} = 2 \parallel 4 + 6 = 22/3$$



iii) $t > 4$



- (1) KCL at node P
- (2) $V_P = 6i + Ldi/dt$

For $t > 0$,

$$i(0^+) = i(0^-) = 0$$

For $0 < t \leq 4$,

$$i(\infty) = \frac{40}{4 + 6} = 4 \text{ (A)}$$

$$\tau = \frac{5}{4 + 6} = 0.5 \text{ (s)}$$

$$i(t) = 4 + (0 - 4)e^{-t/0.5} = 4(1 - e^{-2t}) \text{ (A)}$$

$$i(2) = 4(1 - e^{-2 \times 2}) \approx 3.93 \text{ (A)}$$

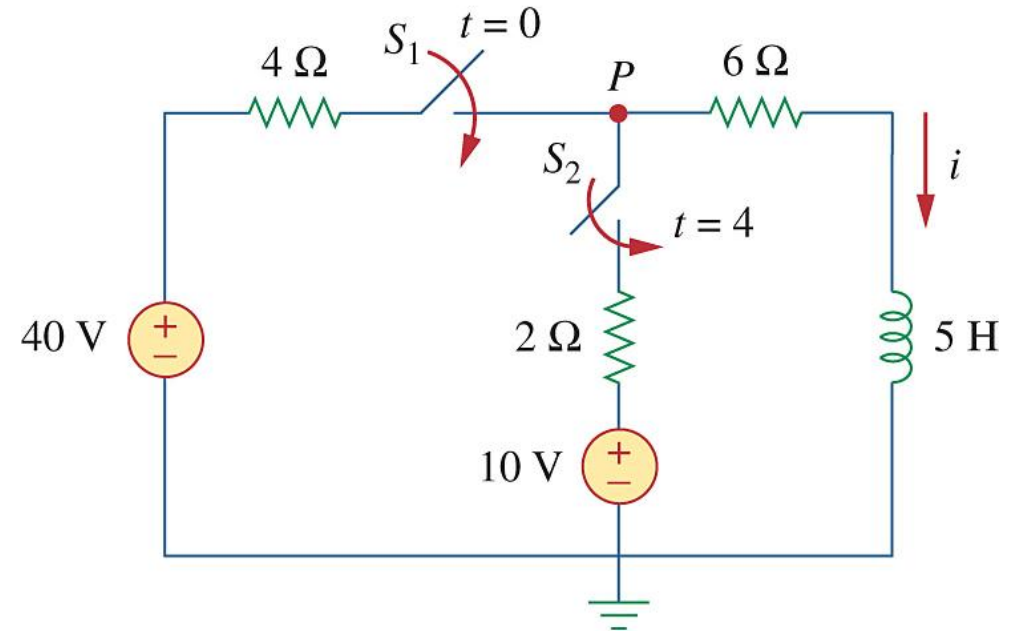


Figure 7.53

For $t > 4$,

$$i(4^+) = i(4^-) = 4(1 - e^{-2 \times 4}) = 4(1 - e^{-8}) \text{ (A)}$$

$$v_P(\infty) = 40 \times \frac{2 \parallel 6}{4 + 2 \parallel 6} + 10 \times \frac{4 \parallel 6}{2 + 4 \parallel 6}$$

$$= \frac{180}{11} \text{ (V)}$$

$$i(\infty) = \frac{v_P(\infty)}{6} = \frac{30}{11} \text{ (A)}$$

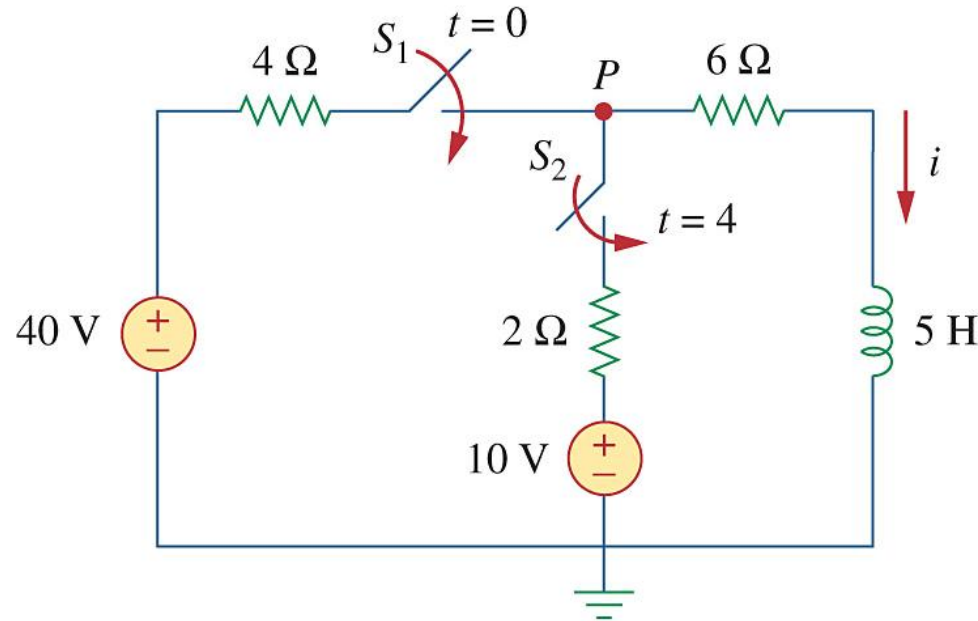


Figure 7.53

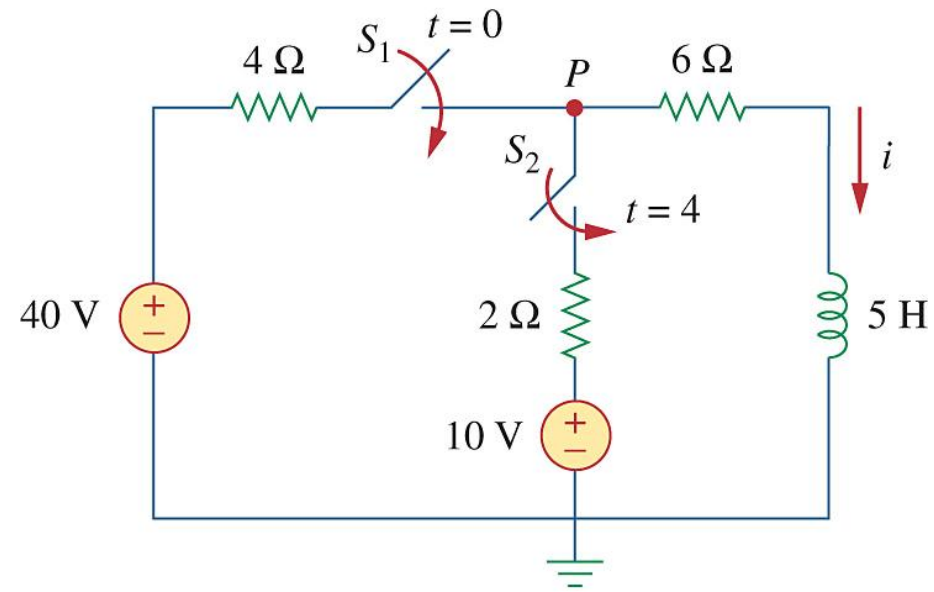


Figure 7.53

$$\tau = \frac{5}{6 + 4 \parallel 2} = \frac{15}{22} \text{ (s)}$$

$$i(t) = \frac{30}{11} + \left(4(1 - e^{-8}) - \frac{30}{11} \right) e^{-\frac{(t-4)}{(15/22)}}$$

Time shift property

$$\approx 2.7273 + 1.2714e^{-1.4667(t-4)} \text{ (A)}$$

$$i(5) = 2.7273 + 1.2714e^{-1.4667(5-4)} \approx 3.02 \text{ (A)}$$

7.7 First-Order Op Amp Circuits

An op amp circuit containing a storage element will exhibit **first-order behavior**. For practical reasons, inductors are hardly ever used in op amp circuits; we will consider Op Amp circuits with RC.

- Possible location of the capacitor
 - (1) At the input
 - (2) At the output
 - (3) At the feedback loop

Example 7.14 For the op amp circuit in Fig. 7.55(a), find v_o for $t > 0$, given that $v(0) = 3$ V. Let $R_f = 80$ k Ω , $R_1 = 20$ k Ω , and $C = 5$ μ F.

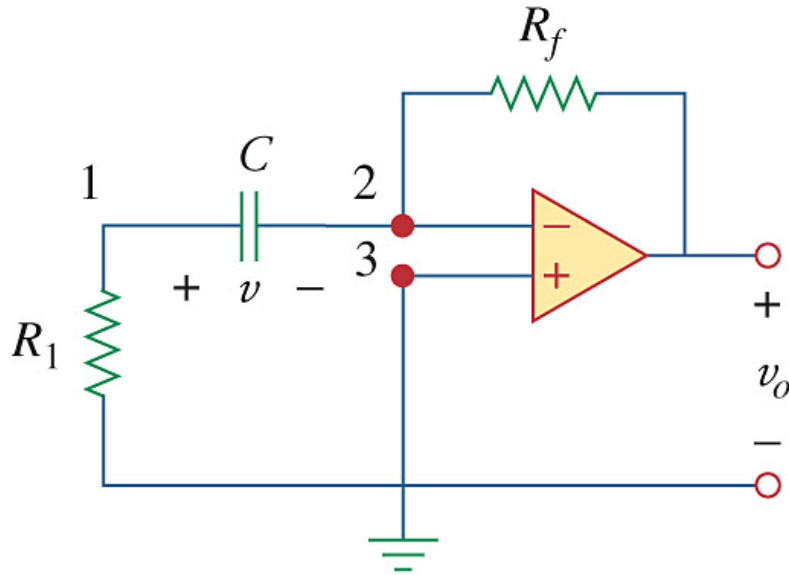
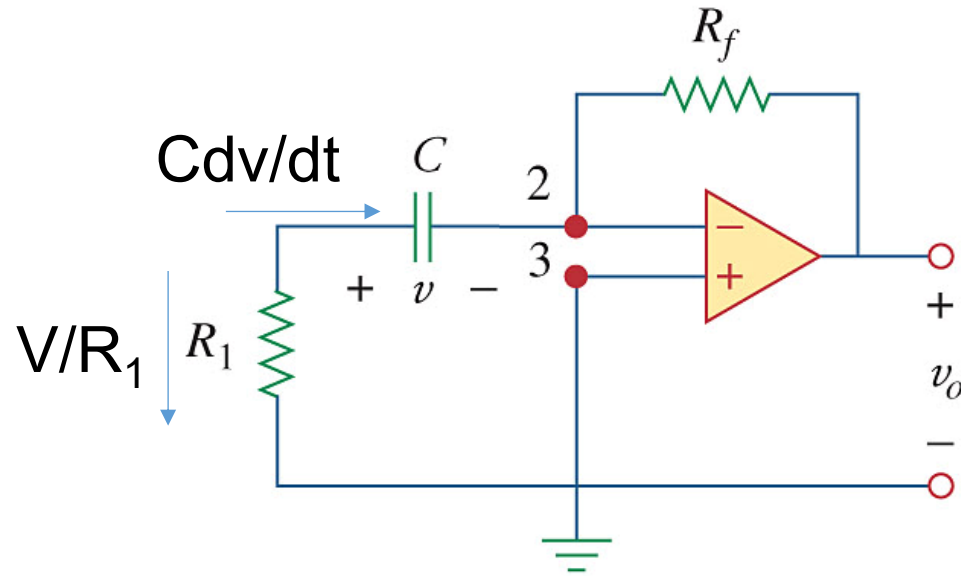


Figure 7.55(a)

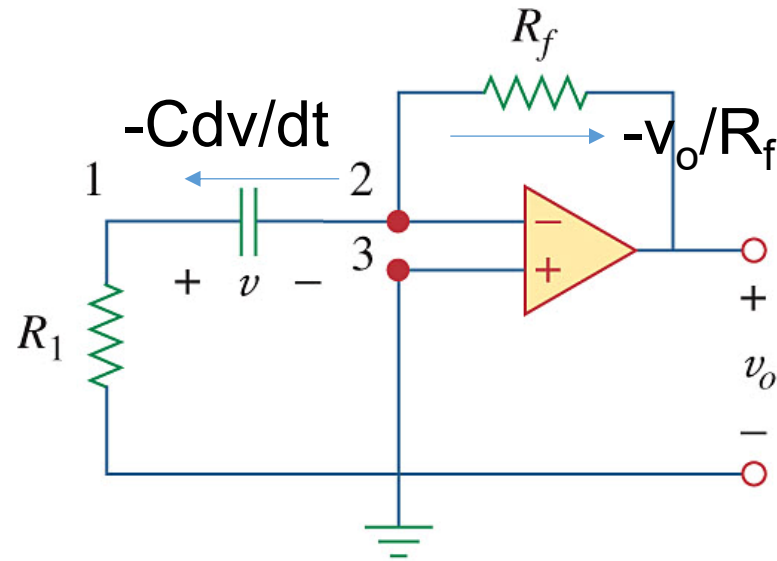


By KCL

$$\frac{V}{R_1} + C \frac{dv}{dt} = 0 \rightarrow \frac{dV}{dt} + \frac{1}{RC_1} v = 0$$

$$\tau = R_1 C = 20 \times 10^3 \times 5 \times 10^{-6} = 0.1 \text{ (s)}$$

$$v = v(0)e^{-t/\tau} = 3e^{-t/0.1} = 3e^{-10t} \text{ (V)}$$



KCL from node 2 gives

$$\begin{aligned}
 -C \frac{dv}{dt} + \frac{0 - v_o}{R_f} &= 0 \\
 \rightarrow v_o = -R_f C \frac{dv}{dt} &= -\left(5 \times 10^{-6} \frac{d}{dt}(3e^{-10t})\right) \times 80 \times 10^3 \\
 &= -0.4 \times \left(3 \times (-10e^{-10t})\right) \\
 &= 12e^{-10t} \text{ (V)}
 \end{aligned}$$

Example 7.15 Determine $v(t)$ and $v_o(t)$ in the circuit of Fig. 7.57.

Solution :

For $t < 0$, **No energy at the circuit**

$$v_1(t) = 0, v_o(t) = 0, v(t) = 0$$

$$v(0^-) = 0$$

For $t > 0$,

$$v_1(t) = 3 \times \frac{20}{10 + 20} = 2 \text{ (V)}$$

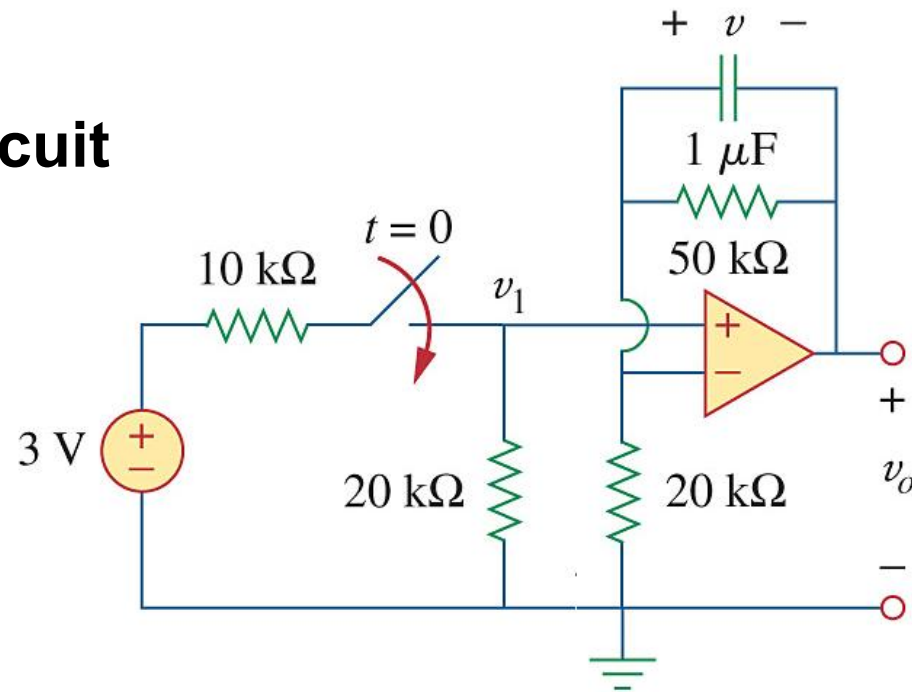
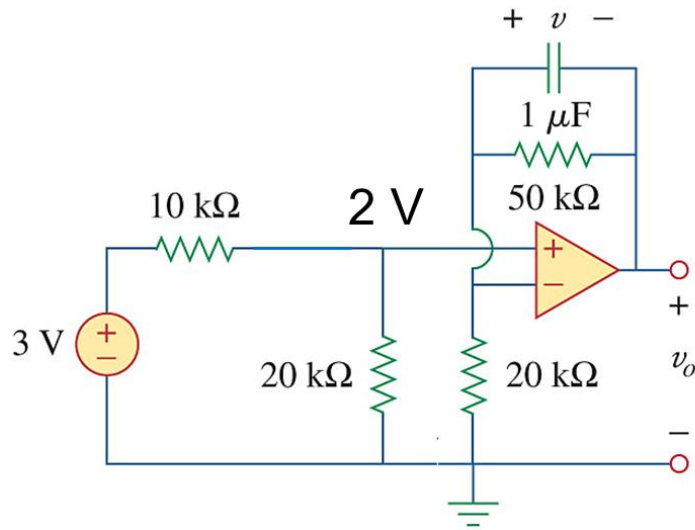
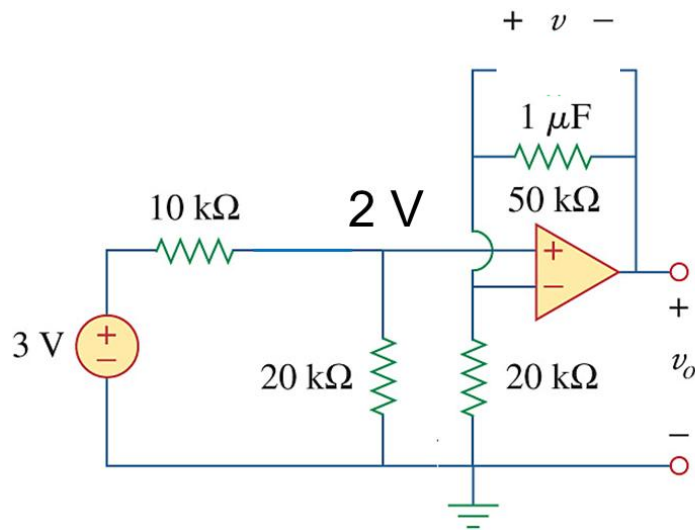


Figure 7.57

(ii) $t > 0$



$$(1) RC = 10^{-6} \times 50k = 1/20$$



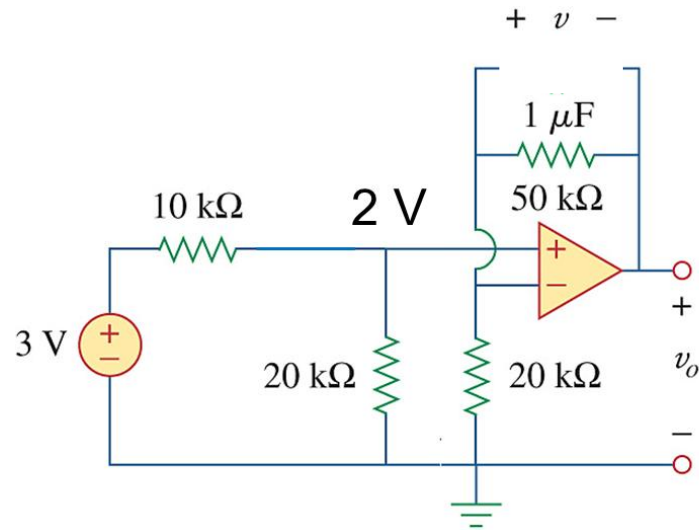
(2) $v(\infty)$

By KCL: $\frac{2}{20k} + \frac{2-v_o}{50k} = 0, v_o = 7 \text{ V}$

Thus $v(\infty) = -5 \text{ V}$

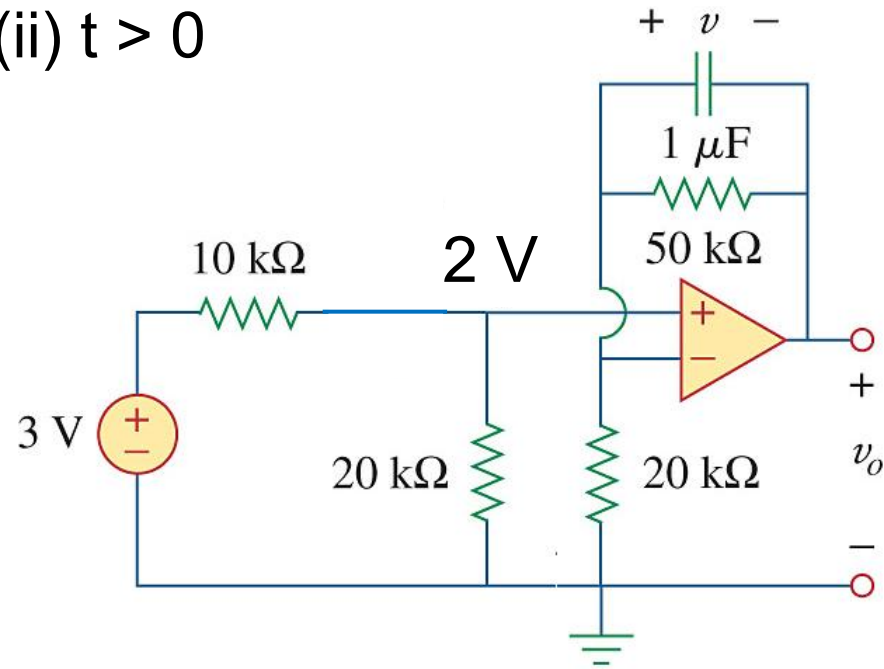
$$v(t) = -5 + [0 + 5]e^{-20t} = 5e^{-20t} - 5$$

(ii) $t > 0$



Because $2 - v_o = v$
 where $v(t) = 5e^{-20t} - 5$ [V]
 $v_o(t) = 7 - 5e^{-20t}$ [V]

(ii) $t > 0$

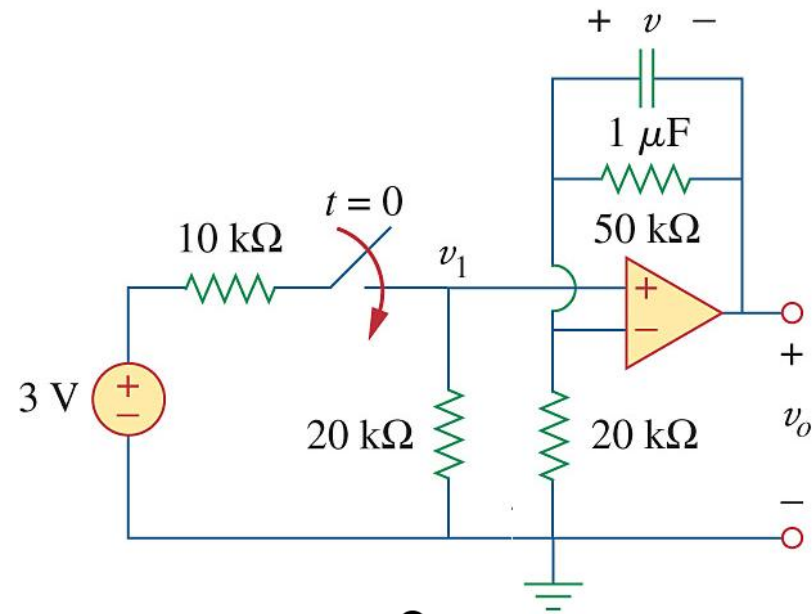


(i) $2 - v_o = v$

(ii) By KVL: $\frac{2}{20k} + \frac{2-v_o}{50k} + C \frac{dv}{dt} = 0$

$\frac{2}{20k} + \frac{v}{50k} + 10^{-6} \frac{dv}{dt} = 0 \times 10^6$

$\frac{dv}{dt} + 20v = -100$



$$(1 \times 10^{-6}) \frac{dv}{dt} + \frac{v}{50 \times 10^3} = -\frac{2}{20 \times 10^3}$$

$$\frac{dv}{dt} + 20v = -100$$

$$v(t) = Ae^{-20t} + B = 5e^{-20t} - 5 \text{ (V)}$$

$$v_o(t) = -v(t) + 2 = 7 - 5e^{-20t} \text{ (V)}$$